Johns Hopkins University

601.455/655 Computer Integrated Surgery I

Assignment 1

Pivot Calibration

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Report for CIS I Assignment 1: Pointer / Pivot Calibration

Contents

1 Introduction **3**

1.1 Problem Statement and Objective 3

1.2 System Components and Coordinate Frames 3

1.3 Ergonomic Considerations 4

2 Mathematical Approach **5**

2.1 Rigid 3D–3D Registration 5

2.2 Marker-Body Geometry Refinement 6

2.3 Pivot Calibration 7

2.4 Stacked Least-Squares Formulation 8

2.4.1 Minimum Data, Rank, and Corner Cases 9

2.4.2 Covariance Estimate 14

2.5 Tip-Position Prediction 14

3 Algorithmic Steps **16**

3.1 Design and Initialization 16

3.2 Marker-Body Refinement 17

3.3 Calibration Data Collection 18

3.4 Pivot Calibration 19

3.5 Test-Pose Validation 20

4 Program Structure **21**

4.1 Functions Used 21

4.1.1 Simulation and Tracking Functions 22

4.1.2 Designed Functions 22

4.2 Program Organization 24

4.3 Key Design Parameters 25

5 Validation Approach **25**

5.1 Unit Tests 25

5.1.1 Frame Transformations and Rigid Registration Tests 25

5.1.2 Marker Geometry Refinement Tests 27

5.1.3 Calibration Pose Generation and Rotation Diversity Tests 28
5.1.4 Tip Calibration Solver Tests 29
5.1.5 Tip Position Prediction Tests 31
5.2 Calibration Validation 32
5.2.1 Marker Geometry Refinement Verification 32
5.2.2 Held-Out Sensor Validation 33
5.3 End-to-End Validation 34
5.3.1 Test Pose Generation and Single-Observation Protocol 34
5.3.2 Error Metrics and Evaluation Criteria 35
5.4 Repeatability and Sensitivity Testing 37
5.4.1 Repeatability Across Simulation Runs 38
5.4.2 Sensitivity to Marker Geometry 38

6 Final Results and Discussion **39**
6.1 Final System Configuration 39
6.2 Unit Test Results 42
6.3 Calibration Results 45
6.4 End-to-End Positioning Accuracy 47
6.5 Design Optimization and Sensitivity Analysis 49
6.6 Error Analysis and Limitations 51

7 Conclusion **55**

A Supplementary Material **56**

1 Introduction

Pivot calibration is a fundamental procedure in Computer-Integrated Surgery (CIS) used to determine the position of a surgical instrument's tip relative to its attached tracking markers. Accurate localization of the instrument tip is essential for image-guided surgical navigation, where the position of an instrument must be determined within a common coordinate system. Although an optical tracking system can measure the position and orientation of a rigid marker body, the physical location of the instrument tip relative to that marker body is not directly known and must be estimated through calibration.

This project focuses on calibrating a tracked pointer using an optical tracking system, a pointer equipped with a rigid marker body, and a calibration object with its own tracking markers. The tracker measures the three-dimensional positions of the markers within its working volume, allowing the poses of the pointer and calibration object to be estimated. The objective is to determine the position of the pointer-tip relative to its marker body and to achieve a calibration accuracy of approximately 2 mm.

The pivot calibration procedure involves collecting multiple observations of the pointer at different orientations while maintaining its tip at a fixed location relative to the calibration object. An initial refinement of the marker-body geometry is performed to improve the accuracy of pose estimation. The collected rigid-body transformations are then used to formulate a linear least-squares problem to estimate the pointer-tip offset. The resulting calibration is evaluated using the consistency of the estimated tip position in observations of the instrument tip at different locations.

1.1 Problem Statement and Objective

The objective of this project is to calibrate a tracked pointer by determining the three-dimensional position of its tip relative to the pointer's marker-body coordinate frame. Although the optical tracking system measures the position and orientation of the pointer's marker body, the tip location is not directly measured. Therefore, a calibration procedure is required to estimate the fixed offset between the pointer tip and its marker-body origin. The offset is primarily because of uncertainity introduced into the system from various factors such as manufacturing tolerances of the marker bodies, noise from sensors measurements affecting pose estimation, and other systematic errors.

The primary goal is to achieve a pointer-tip localization accuracy of approximately **2 mm** within the tracking workspace. This is accomplished by collecting multiple measurements of the pointer and calibration object at different orientations and solving a linear least-squares problem to estimate the tip offset. Refinement of the nominal marker-body geometry is also performed to improve the reliability of the estimated rigid-body transformations.

1.2 System Components and Coordinate Frames

The experimental setup consists of an optical tracker, an instrument tip (also referred to as a pointer), and a calibration object. We define coordinates frames locally to the rigid marker body equipped onto both the instrument and the calibration object.

The system uses the following coordinate frames:

1. **Tracker Frame :** The global reference coordinate frame established by the optical tracking system. The measured positions and estimated poses of all tracked marker bodies are expressed relative to this frame.

2. **Pointer Frame :** A local coordinate frame rigidly attached to the pointer marker body. The position of the pointer-tip, denoted by $\mathbf{p}_{tip}^{P}$, is expressed in this frame and is the primary unknown determined by calibration.

3. **Calibration Frame :** A local coordinate frame attached to the calibration object's marker body. It provides a reference for the fixed pivot location during calibration.

4. **Reference Body Frame :** A local coordinate frame rigidly attached to an additional reference marker body is used during the validation stage. The optical tracker measures the pose of the reference body, allowing the rigid-body transformation $\mathbf{F}^{-1}$ to be estimated.

The primary physical components of the system include:

- **Pointer Tip ($\mathbf{p}_{tip}^{Tracker}$) :** The physical endpoint of the tracked pointer. During pivot calibration, the tip is maintained at a fixed location while the pointer undergoes different orientations. The position of the pointer tip with respect to the tracker coordinate frame is determined using the rigid-body transformation:

$$\mathbf{p}_{tip}^{Tracker} = \mathbf{F}_{ptr}\mathbf{p}_{ptr}^{PointerFrame} \tag{1}$$

  where $\mathbf{F}_{ptr}$ represents the rigid-body transformation from the tracker frame to the pointer frame, and $\mathbf{p}_{ptr}^{PointerFrame}$ represents the fixed position of the pointer tip relative to the pointer marker-body frame. For convenience, the superscript will be dropped.

- **Marker Bodies ($\mathbf{a}_{ptr}^{\text{Pointer Frame}}$, $\mathbf{a}_{cal}^{\text{Calibration Frame}}$):** Rigid bodies attached to the pointer and calibration object for optical tracking. Their measured marker positions will be used to estimate the corresponding rigid-body transformations.

1.3 Ergonomic Considerations

Beyond geometric accuracy, the marker-body designs incorporate practical considerations related to ergonomics, visibility, and ease of operation. The pointer marker body is intended to maintain a compact, lightweight configuration that facilitates controlled manipulation while minimizing hand obstruction and marker occlusion. Similarly, the calibrator marker body uses a spatially distributed marker arrangement to support reliable optical tracking, while its supporting structure is intended to provide mechanical stability and accessibility during calibration. These considerations highlight the trade-off between tracking performance and practical usability in computer-integrated surgical systems. See the appendix for 3D sketches of marker body and calibration object.

2 Mathematical Approach

2.1 Rigid 3D–3D Registration

The tracker utilizes SVD/Arun-style registration to do 3D-3D point registration between the observed marker point locations and the known marker locations in the body local frame. This is implemented using `procrustes_solver()`.

The tracker estimates the pose of a marker body by registering its nominal marker positions $\mathbf{a}_{ptr,k}$, expressed in the local pointer frame, with the corresponding observed marker positions $\mathbf{b}_{ptr,k}$, expressed in the tracker frame.

The rigid transformation $F^t_{ptr} = (R, t)$ maps points from the pointer frame to the tracker frame:

$$\mathbf{b}_{ptr,k} \approx R\mathbf{a}_{ptr,k} + t \tag{2}$$

Due to measurement noise and manufacturing inaccuracies, the tracker estimates the best-fit rotation and translation by minimizing the sum of squared registration residuals:

$$\boxed{(\hat{R}, \hat{t}) = \underset{R\in SO(3),\, t\in\mathbb{R}^3}{\operatorname{argmin}} \sum_{k=1}^{K} \|\mathbf{b}_{ptr,k} - (R\mathbf{a}_{ptr,k} + t)\|^2} \tag{3}$$

This is a constrained least-squares problem since the rotation must satisfy $R^T R = I$ and $\det(R) = 1$.

The tracker solves this problem using Arun's SVD-based registration method. First, the centroids are calculated:

$$\bar{\mathbf{a}}_{ptr} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{a}_{ptr,k}, \qquad \bar{\mathbf{b}}_{ptr} = \frac{1}{K}\sum_{k=1}^{K} \mathbf{b}_{ptr,k} \tag{4}$$

The centered points are used to construct the cross-covariance matrix:

$$H = \sum_{k=1}^{K} (\mathbf{a}_{ptr,k} - \bar{\mathbf{a}}_{ptr})(\mathbf{b}_{ptr,k} - \bar{\mathbf{b}}_{ptr})^T \tag{5}$$

Applying Singular Value Decomposition:

$$H = U\Sigma V^T \tag{6}$$

The optimal rotation is:

$$\hat{R} = V \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & \det(VU^T) \end{bmatrix} U^T \tag{7}$$

The translation is recovered from the centroids:

$$\hat{t} = \bar{\mathbf{b}}_{ptr} - \hat{R}\bar{\mathbf{a}}_{ptr} \tag{8}$$

Finally, the estimated rigid-body transformation is:

$$\boxed{\hat{F}_{ptr}^{Tracker} = \begin{bmatrix} \hat{R} & \hat{t} \\ 0 & 1 \end{bmatrix}} \tag{9}$$

2.2 Marker-Body Geometry Refinement

The nominal marker positions of a rigid marker body may differ from their actual physical locations due to manufacturing tolerances, assembly imperfections, and measurement uncertainties. Since the tracker estimates rigid-body transformations using 3D–3D registration, inaccuracies in the nominal marker geometry can introduce errors in the estimated pose. To improve the accuracy of subsequent pose estimation and pivot calibration, the nominal marker coordinates are refined using repeated tracker observations.

For each observation j, the tracker estimates the rigid-body transformation $F_{ptr,j}^{t} = (R_j, t_j)$ using the SVD-based registration method described previously. The observed marker positions $\mathbf{b}_{ptr,k}^{(j)}$, expressed in the tracker frame, are transformed back into the pointer's local coordinate frame using the inverse of the estimated rigid-body transformation:

$$\boxed{(F_{ptr,j}^{t})^{-1} = \begin{bmatrix} R_j^T & -R_j^T t_j \\ 0 & 1 \end{bmatrix}} \tag{10}$$

$$\boxed{\mathbf{a}_{ptr,k}^{(j)} = \left(F_{ptr,j}^{t}\right)^{-1} \cdot \mathbf{b}_{ptr,k}^{(j)}} \tag{11}$$

where $\mathbf{a}_{ptr,k}^{(j)}$ represents the estimated local position of marker k obtained from observation j.

$$\mathbf{a}_{ptr,k}^{(j)} = R_j^T \left(\mathbf{b}_{ptr,k}^{(j)} - t_j\right) \tag{12}$$

where $\mathbf{a}_{ptr,k}^{(j)}$ represents the estimated local position of marker k obtained from observation j.

Multiple observations are collected to reduce the effects of measurement noise. The estimated local coordinates of each marker are averaged across N observations to obtain an updated marker position:

$$\boxed{\bar{\mathbf{a}}_{ptr,k} = \frac{1}{N} \sum_{j=1}^{N} \mathbf{a}_{ptr,k}^{(j)}} \tag{13}$$

where N represents the number of observations and $\bar{\mathbf{a}}_{ptr,k}$ denotes the refined local coordinates of marker k.

The refinement is performed iteratively. Following each iteration, the nominal marker coordinates are replaced with the averaged coordinates, which are subsequently used to estimate the rigid-body transformations in the next iteration.

In this project, three refinement iterations are performed, with 30 tracker observations collected per iteration. The resulting marker-body geometry is used for subsequent pose estimation and pivot calibration.

2.3 Pivot Calibration

The objective of pointer-tip calibration is to determine the fixed three-dimensional position of the pointer tip, $\mathbf{p}_{tip}$, relative to the pointer's local marker-body coordinate frame. The implemented approach uses calibrated sensor measurements to determine the tip location rather than estimating an unknown stationary pivot point.

For each observation j, the tracker estimates the rigid-body transformations of the pointer and calibration object, denoted by $F_{ptr,j}$ and $F_{cal,j}$, respectively. The calibrated sensor provides the measured position $\mathbf{p}_{sensed,j}$, expressed in the calibration object's local coordinate frame.

Since the pointer-tip position calculated from the pointer frame and the sensor-measured position describe the same physical point, their coordinates in the tracker frame must be equal:

$$\boxed{R_{ptr,j}\mathbf{p}_{tip} + t_{ptr,j} = R_{cal,j}\mathbf{p}_{sensed,j} + t_{cal,j}} \tag{14}$$

Rearranging to isolate the unknown pointer-tip position:

$$R_{ptr,j}\mathbf{p}_{tip} = R_{cal,j}\mathbf{p}_{sensed,j} + t_{cal,j} - t_{ptr,j} \tag{15}$$

This can be expressed as a linear system:

$$\boxed{A_j\mathbf{p}_{tip} = \mathbf{b}_j} \tag{16}$$

where

$$A_j = R_{ptr,j} \tag{17}$$

and

$$\mathbf{b}_j = R_{cal,j}\mathbf{p}_{sensed,j} + t_{cal,j} - t_{ptr,j} \tag{18}$$

Each observation provides three linear equations for the three unknown coordinates of the pointer tip. To improve the reliability of the estimate, N observations are collected and stacked into an overdetermined linear system:

$$\begin{bmatrix} R_{ptr,1} \\ R_{ptr,2} \\ \vdots \\ R_{ptr,N} \end{bmatrix} \mathbf{p}_{tip} \approx \begin{bmatrix} R_{cal,1}\mathbf{p}_{sensed,1} + t_{cal,1} - t_{ptr,1} \\ R_{cal,2}\mathbf{p}_{sensed,2} + t_{cal,2} - t_{ptr,2} \\ \vdots \\ R_{cal,N}\mathbf{p}_{sensed,N} + t_{cal,N} - t_{ptr,N} \end{bmatrix} \tag{19}$$

where

$$\mathbf{A} = \begin{bmatrix} R_{ptr,1} \\ R_{ptr,2} \\ \vdots \\ R_{ptr,N} \end{bmatrix} \in \mathbb{R}^{3N\times 3}, \qquad \mathbf{P}_{tip} = \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} \in \mathbb{R}^3, \qquad \mathbf{b} = \begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \vdots \\ \mathbf{b}_N \end{bmatrix} \in \mathbb{R}^{3N} \tag{20}$$

Such that

$$\boxed{A\mathbf{p}_{tip} = \mathbf{b}} \tag{21}$$

An exact solution may not exist for $\mathbf{p_{tip}}$ simply by doing A^{-1}*$\mathbf{b}$ since A is a $\nVdash\mathbb{N} \times \nVdash$ matrix and not a square matrix it is generally not invertible. Therefore, the pointer-tip coordinates are estimated by minimizing the sum of squared residuals to find calibrated pointer-tip value:

$$\boxed{\hat{\mathbf{p}}_{tip} = \underset{\mathbf{p}_{tip}}{\operatorname{argmin}} \|A_j\mathbf{p}_{tip} - \mathbf{b}_j\|_2^2} \tag{22}$$

We will look into this in more depth in the next section

2.4 Stacked Least-Squares Formulation

For each observation j, the calibration residual is defined as the difference between the true pointer-tip position and the sensor-measured position of the pointer tip:

$$\mathbf{r}_j = R_{ptr,j}\mathbf{p}_{tip} - (R_{cal,j} * \mathbf{p_{sensed,j}} + t_{cal,j} - t_{ptr,j}) \tag{23}$$

We use the previous definitions for A_j , $\mathbf{b}_j$ to rewrite above equation as:

$$\boxed{\mathbf{r}_j = A_j\mathbf{p}_{tip} - \mathbf{b}_j} \tag{24}$$

To account for measurement uncertainties, residuals from N observations are stacked into a single system:

$$\underbrace{\begin{bmatrix} \mathbf{r}_1 \\ \mathbf{r}_2 \\ \vdots \\ \mathbf{r}_N \end{bmatrix}}_{\mathbf{r}} = \underbrace{\begin{bmatrix} A_1 \\ A_2 \\ \vdots \\ A_N \end{bmatrix}}_{A} \mathbf{p}_{tip} - \underbrace{\begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \\ \vdots \\ \mathbf{b}_N \end{bmatrix}}_{\mathbf{b}} \tag{25}$$

Therefore,

$$\boxed{\mathbf{r} = A\mathbf{p}_{tip} - \mathbf{b}} \qquad A \in \mathbb{R}^{3N\times 3}, \quad \mathbf{p}_{tip} \in \mathbb{R}^3, \quad \mathbf{b} \in \mathbb{R}^{3N} \tag{26}$$

The linear least-squares problem is formulated by minimizing the sum of squared residuals:

$$\boxed{\hat{\mathbf{p}}_{tip} = \underset{\mathbf{p}_{tip}}{\operatorname{argmin}} \sum_{j=1}^{N} \|\mathbf{r}_j\|_2^2 = \underset{\mathbf{p}_{tip}}{\operatorname{argmin}} \|A\mathbf{p}_{tip} - \mathbf{b}\|_2^2} \tag{27}$$

Assuming A has full column rank, the least-squares solution is:

$$\boxed{\hat{\mathbf{p}}_{tip} = (A^T A)^{-1} A^T \mathbf{b}} \tag{28}$$

This solution provides the estimated three-dimensional coordinates of the pointer tip relative to its local marker-body frame.

2.4.1 Minimum Data, Rank, and Corner Cases

Referring to the residual linear equation:

$$\mathbf{r} = A\mathbf{p}_{tip} - \mathbf{b} \tag{29}$$

The unknown pointer-tip position $\mathbf{p}_{tip} \in \mathbb{R}^3$ contains three coordinates, while $A \in \mathbb{R}^{3N\times 3}$. Each observation provides three equations. So for more than $N \geq 1$ we have N *3 equations and 3 unknowns. Although one valid observation is mathematically sufficient, multiple observations improve reliability and allow residual-based uncertainty estimation.

We also need to check if the least squares solution can be obtained which depends on the Rank of A as follows:

$$\hat{\mathbf{p}}_{tip} = (A^T A)^{-1} A^T \mathbf{b} \tag{30}$$

$(A^T A)$ needs to be invertible. Since A consists of stacked rotation matrices:

$$A^T A = \sum_{j=1}^{N} R_{ptr,j}^T R_{ptr,j} = N I_3 \tag{31}$$

For $N \geq 1$, the matrix A has full column rank, ensuring an unique least-squares solution.

Its determinant is:

$$\det(A^T A) = \det(N I_3) \tag{32}$$

For a 3×3 identity matrix:

$$\det(NI_3) = N^3 \tag{33}$$

Since $N \geq 1$:

$$N^3 \neq 0 \tag{34}$$

Also note A is a $3X3$ matrix in $SO(3)$ it has three linearly independent columns. Hence:

$$\text{rank}(A) = 3 \tag{35}$$

A unique least-squares solution requires $N \geq 1$ observations and $rank(A) = 3$ to be solvable mathematically

Unlike conventional pivot calibration, identical or nearly identical orientations do not cause rank degeneracy in this sensor-based formulation. We conclude the linear least square formulation is not vulnerable to orientation-induced rank deficiency because each observation contributes a full-rank relative rotation matrix. Thus mathematical solvability is not the concern and we need to check for conditioning

Now suppose we perturb the observation measurements:

$$\mathbf{b}_{\text{measured}} = \mathbf{b}_{\text{true}} + \Delta\mathbf{b}_{perturbtion} \tag{36}$$

The estimated tip position becomes:

$$\hat{\mathbf{p}}_{tip} + \Delta\mathbf{p}_{tip} = A^{+}(\mathbf{b} + \Delta\mathbf{b}) \tag{37}$$

Note $\mathbf{A}^{+}$ here is the pseudoinverse

We obtain:

$$\Delta\mathbf{p}_{tip} = A^{+}\Delta\mathbf{b} \tag{38}$$

Using equation (33) we can rewrite as:

$$\boxed{\Delta\mathbf{p}_{tip} = \frac{1}{N}A^{T}\Delta\mathbf{b}} \tag{39}$$

Here the error amplification term is rotation matrix A which will be used to discuss conditioning of the system. Numerical stability is assessed using the condition number which measures the difference between the maximum and minimum stretching produced by A:

$$\boxed{\kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)} = 1} \tag{40}$$

Here $\sigma(A)$ is the singular value of A. We use SVD on $A^TA = V\Sigma^T\Sigma V^T$ We obtain the following relation here

$$\sigma_i = \sqrt{\lambda_i} \tag{41}$$

We already established in (33):

$$A^T A = N I_3 \tag{42}$$

Therefore:

$$A^T A = \begin{bmatrix} N & 0 & 0 \\ 0 & N & 0 \\ 0 & 0 & N \end{bmatrix} \tag{43}$$

Its eigenvalues are:

$$\lambda_1 = \lambda_2 = \lambda_3 = N \tag{44}$$

Since $\lambda_i = \sigma_i^2$:

$$\sigma_i^2 = N \tag{45}$$

Taking the square root:

$$\boxed{\sigma_i = \sqrt{N}} \quad \sigma_{\max}(A) = \sqrt{N}, \quad \sigma_{\min}(A) = \sqrt{N} \tag{46}$$

Therefore:

$$\boxed{\kappa_2(A) = 1} \tag{47}$$

The corner cases of pointer-tip calibration can be analyzed using the rank, singular values, and condition number of the stacked measurement matrix A we discussed previously.

Case 1: Single Observation

For $N = 1$, the calibration system contains three equations and three unknowns:

$$A_1 \mathbf{p}_{tip} = \mathbf{b}_1 \tag{48}$$

Since A_1 is orthogonal:

$$\text{rank}(A_1) = 3, \qquad \kappa_2(A_1) = 1 \tag{49}$$

The unique solution is:

$$\mathbf{p}_{tip} = A_1^T \mathbf{b}_1 \tag{50}$$

Case 2: Multiple Identical Orientations

If the relative orientation remains constant:

$$A_1 = A_2 = \cdots = A_N = Q, \qquad Q \in SO(3) \tag{51}$$

Then:

$$A = \begin{bmatrix} Q \\ Q \\ \vdots \\ Q \end{bmatrix} \tag{52}$$

The normal matrix becomes:

$$A^T A = N Q^T Q = N I \tag{53}$$

Therefore:

$$\text{rank}(A) = 3, \qquad \kappa_2(A) = 1 \tag{54}$$

The system remains uniquely solvable, although repeated orientations may prevent orientation-dependent systematic errors from being identified.

Case 3: Multiple Varied Orientations

For observations collected at different relative orientations:

$$A_i \neq A_j \tag{55}$$

where each A_j remains orthogonal. Therefore:

$$A^T A = \sum_{j=1}^{N} A_j^T A_j = N I \tag{56}$$

and:

$$\text{rank}(A) = 3, \qquad \kappa_2(A) = 1 \tag{57}$$

Orientation diversity does not change the theoretical conditioning of the linear system, but can help identify orientation-dependent measurement biases.

Case 4: No Valid Observations

For $N = 0$, the measurement matrix contains no rows:

$$A \in \mathbb{R}^{0 \times 3} \tag{58}$$

Therefore:

$$\text{rank}(A) = 0 and A^T A = 0_{3\times 3} \tag{59}$$

The matrix $A^T A$ is singular, and a unique calibration solution cannot be obtained.

Case 5: Invalid or Non-orthogonal Pose Matrices

If the estimated rotation matrices do not satisfy:

$$R^T R = I \tag{60}$$

then the observation matrices may also be non-orthogonal:

$$A_j^T A_j \neq I \tag{61}$$

Consequently:

$$A^T A \neq NI \tag{62}$$

The condition number is no longer guaranteed to equal one:

$$\kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)} \geq 1 \tag{63}$$

The system is rank deficient too for this case and its condition number is infinite.

Case 6: Correlated Measurement Errors

Consider the noisy measurement model with noise $\boldsymbol{w}$:

$$\mathbf{b} = A\mathbf{p}_{tip} + \boldsymbol{w} \tag{64}$$

where:

$$E[\boldsymbol{w}] = 0, \qquad \text{Cov}(\boldsymbol{w}) = \Sigma \tag{65}$$

For independent and isotropic measurement errors:

$$\Sigma = \sigma^2 I \tag{66}$$

The covariance of the least-squares estimate is:

$$\text{Cov}(\hat{\mathbf{p}}_{tip}) = \sigma^2 (A^T A)^{-1} = \frac{\sigma^2}{N} I \tag{67}$$

For correlated measurement errors:

$$\Sigma \neq \sigma^2 I \tag{68}$$

The covariance of the ordinary least-squares estimate becomes:

$$\mathrm{Cov}(\hat{\mathbf{p}}_{tip}) = (A^T A)^{-1} A^T \Sigma A (A^T A)^{-1} \tag{69}$$

Therefore, correlated observations may provide less uncertainty reduction than statistically independent measurements.

2.4.2 Covariance Estimate

From the residual vector derived in equation (28) we can estimate the residual-based variance by dividing the sum of squared residuals by the remaining degrees of freedom

$$\boxed{\hat{\sigma}^2 = \frac{\|\mathbf{r}\|_2^2}{3N - 3}} \tag{70}$$

where N is the number of calibration observations. Each observation contributes three scalar equations, resulting in $3N$ measurements. Since three unknown pointer-tip coordinates are estimated, the residual degrees of freedom are $3N - 3$.

From the equation (74) for general case of correlated measurement errors we can derive the covariance of the estimated pointer-tip position. We assume - that measurement errors are independent, have zero mean, and possess equal variance in all coordinate directions - as follows:

$$E[\boldsymbol{w}] = 0, \qquad \mathrm{Cov}(\boldsymbol{w}) = \sigma^2 I \tag{71}$$

we obtain:

$$\boxed{\mathrm{Cov}(\hat{\mathbf{p}}_{tip}) = \sigma^2 (A^T A)^{-1}} \tag{72}$$

The resulting 3×3 covariance matrix describes the estimated uncertainty in the calibrated pointer-tip coordinates. The diagonal elements represent the variances along the three coordinate axes, while the off-diagonal elements represent correlations between the estimated coordinates.

In practice, errors in marker-body geometry, tracker registration, and calibrated sensor measurements may violate these assumptions. Consequently, the covariance estimate characterizes uncertainty under the assumed measurement model but does not fully capture systematic calibration errors or guarantee absolute tip-position accuracy.

2.5 Tip-Position Prediction

Following pointer-tip calibration, the estimated tip position $\hat{\mathbf{p}}_{tip}$ is expressed in the pointer's local marker-body coordinate frame. To determine the tip position within the tracking workspace, the calibrated coordinates are transformed using the observed rigid-body transformation of the pointer.

For each validation observation j, the tracker estimates the pointer transformation $F_{ptr,j} = (R_{ptr,j}, t_{ptr,j})$. The calibrated pointer-tip position is mapped into the tracker coordinate frame as:

$$\mathbf{p}_{tip,j} = R_{ptr,j}\hat{\mathbf{p}}_{tip} + t_{ptr,j} \tag{73}$$

In Step 8, an additional nearby reference marker body is used to evaluate the pointer-tip position. The tracker estimates the reference-body transformation $F_{ref,j} = (R_{ref,j}, t_{ref,j})$, which maps coordinates from the reference-body frame to the tracker frame.

To express the predicted pointer-tip position relative to the reference body, the inverse reference-body transformation is applied:

$$F_{ref,j}^{-1} = \begin{bmatrix} R_{ref,j}^T & -R_{ref,j}^T t_{ref,j} \\ 0 & 1 \end{bmatrix} \tag{74}$$

The relative transformation from the pointer frame to the reference-body frame is:

$$F_{ptr,j}^{ref} = F_{ref,j}^{-1} F_{ptr,j} \tag{75}$$

Applying this relative transformation gives the predicted pointer-tip position in reference-body coordinates:

$$\boxed{\mathbf{p}_{tip,j}^{ref} = R_{ref,j}^T \left(R_{ptr,j}\hat{\mathbf{p}}_{tip} + t_{ptr,j} - t_{ref,j}\right)} \tag{76}$$

The predicted tip position is subsequently compared against the independently known reference tip location, $\mathbf{p}_{true}^{ref}$, using the Euclidean distance:

$$\boxed{e_j = \left\| \mathbf{p}_{tip,j}^{ref} - \mathbf{p}_{true}^{ref} \right\|_2} \tag{77}$$

This procedure evaluates the accuracy of the calibrated pointer-tip position across multiple validation observations.

3 Algorithmic Steps

3.1 Design and Initialization

Design and Initialization

1: Define four non-coplanar marker positions for the pointer body:
2: $a_{ptr} = \begin{Bmatrix} (0,0,0),\ (50,0,0), \\ (0,80,0),\ (0,0,60) \end{Bmatrix}$ mm
3: Define approximate pointer tip position:
4: $p_{tip,approx} = (0,0,-150)$ mm
5: Define four non-coplanar marker positions for the calibration body:
6: $a_{cal} = \begin{Bmatrix} (400,-200,0),\ (400,200,0), \\ (-400,-200,0),\ (-400,200,100) \end{Bmatrix}$ mm
7: Define approximate tracker Frame:
8: $F_{tracker} = (I, (-1100, 300, 300))$
9: Define approximate calibration-object Frame:
10: $F_{cal} = (I, (650, -200, 300))$
11: Create pointer and calibration marker bodies
12: Place calibration body at its approximate Frame
13: Initialize pointer body at the identity Frame
14: Create pointer using approximate tip position
15: Initialize optical tracker using approximate tracker Frame
16: **return** Initialized pointer, calibration body, and tracker

3.2 Marker-Body Refinement

Marker-Body Refinement

Input: Tracker, marker body, 30 readings, 3 passes
Output: Refined nominal marker positions
for each marker body (pointer and calibrator) **do**
    for each refinement pass (3 passes) **do**
        Initialize empty list of marker-position estimates
        for each tracker reading (30 readings) **do**
            Collect tracker observations of the marker body
            Estimate the body Frame from the tracker reading
            Compute the inverse of the estimated Frame
            for each observed marker **do**
                Obtain marker position in tracker coordinates
                Transform marker position into body coordinates:
                    $a_k = F^{-1} b_k$
                Store transformed marker position
            end for
            Store all marker positions from current reading
        end for
        Stack marker-position estimates from all 30 readings
        for each marker **do**
            Compute mean position across all readings
        end for
        Update nominal marker positions with computed means
    end for
end for
return Refined marker positions

3.3 Calibration Data Collection

Calibration Data Collection

Input: Pointer, calibrator, tracker
Output: Calibration observations
Set required observations to 30
Set minimum orientation separation to 25°
Initialize empty lists for accepted orientations and observations
for each required calibration observation **do**
    while observation has not been accepted **do**
        Generate random XYZ rotation angles between $-70°$ and $70°$
        Construct desired pointer orientation from generated angles
        Assume desired orientation is acceptable
        for each previously accepted orientation **do**
            Compute relative rotation between desired and accepted orientation
            Calculate angular separation using relative rotation matrix
            if angular separation $< 25°$ **then**
                Reject desired orientation
                break
            end if
        end for
        if desired orientation is rejected **then**
            Generate another orientation
            continue
        end if
        Hand-guide pointer near calibrator at desired orientation
        Collect one simultaneous tracker observation of both bodies
        Measure pointer tip using calibration sensor
        if tip is outside sensor range **then**
            Reject observation
            continue
        end if
        Store observed pointer Frame $F_{ptr,j}$
        Store observed calibration Frame $F_{cal,j}$
        Store sensed tip position $p_{sensed,j}$
        Store accepted orientation
        Exit current observation loop
    end while
end for
return F_{ptr}, F_{cal}, and p_{sensed} observation lists

3.4 Pivot Calibration

Pivot Calibration Using Linear Least Squares

Input: Pointer Frames, calibrator Frames, sensed tip positions
Output: Calibrated tip position and covariance
Initialize empty lists for matrix A and vector b
for each calibration observation j **do**
    if sensed tip measurement is unavailable **then**
        Skip observation
        continue
    end if
    Extract pointer rotation $R_{ptr,j}$ and translation $t_{ptr,j}$
    Extract calibrator rotation $R_{cal,j}$ and translation $t_{cal,j}$
    Obtain sensed tip position $p_{sensed,j}$
    Construct observation matrix:
        $A_j = R_{ptr,j}$
    Construct observation vector:
        $b_j = R_{cal,j} p_{sensed,j} + t_{cal,j} - t_{ptr,j}$
    Append A_j and b_j to corresponding lists
end for
if no valid observations exist **then**
    Terminate calibration with an error
end if
Vertically stack all A_j matrices into A
Stack all b_j vectors into b
Solve $Ax \approx b$ using NumPy linear least squares
Extract calibrated tip position p_{tip} from solution x
Compute residual vector:
    $r = Ax - b$
Calculate degrees of freedom:
    $d = \text{length}(b) - \text{rank}(A)$
if $d > 0$ **then**
    Estimate residual variance:
        $\hat{\sigma}^2 = \dfrac{r^T r}{d}$
    Compute estimated tip-position covariance:
        $\text{Cov}(p_{tip}) = \hat{\sigma}^2 (A^T A)^+$
else
    Set covariance entries to undefined values
end if
return p_{tip} and $\text{Cov}(p_{tip})$

3.5 Test-Pose Validation

Test-Pose Validation

Input: Calibrated tip position, tracker, pointer
Output: Tip-position errors across test poses
Define additional nearby reference marker body
Place reference body near the test-pose workspace
Refine reference-body marker positions using Algorithm 2
Define test rotation angles:
    $\theta = \{0, \pm\pi/6, \pm\pi/3, \pm\pi/2\}$
Define translation offsets:
    $t = \{0, \pm100, \pm200\}$ mm
Initialize empty list of candidate test poses
for each combination of XYZ rotation angles **do**
    for each combination of XYZ translation offsets **do**
        Construct rotation matrix from selected angles
        Construct translation around $(300, 300, 300)$ mm
        Combine rotation and translation into test Frame
        Store candidate test Frame
    end for
end for
Randomly select 50 unique test Frames
Initialize empty list of test errors
for each selected test Frame **do**
    Place pointer at desired Frame with precise placement
    Collect **ONE** simultaneous tracker observation
    Extract observed pointer Frame F_{ptr}
    Extract observed nearby reference Frame F_{near}
    Predict tip position relative to nearby reference:
        $p_{pred} = F_{near}^{-1} F_{ptr} p_{tip}$
    Obtain actual Frames and tip position for validation only
    Compute actual tip position in reference coordinates:
        $p_{actual} = F_{near,true}^{-1} F_{ptr,true} p_{tip,true}$
    Calculate Euclidean tip-position error:
        $e = \|p_{pred} - p_{actual}\|$
    Store error for current test pose
end for
Calculate mean error, maximum error, and RMSE
Calculate 95th-percentile error
Count test poses with error ≤ 2 mm
Generate histogram of tip-position errors
return Validation errors and performance metrics

4 Program Structure

4.1 Functions Used

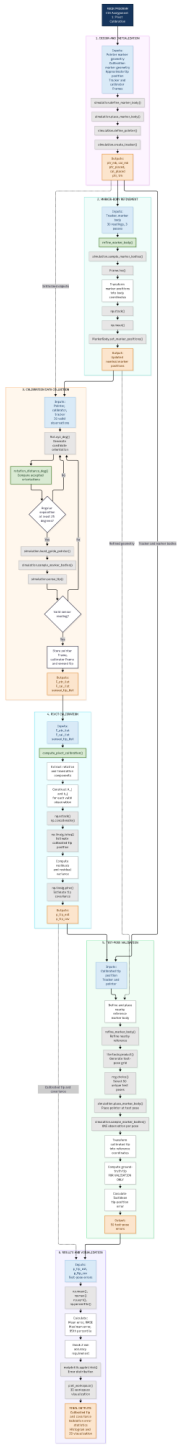


Figure 1: Program structure and function hierarchy for pivot calibration.

4.1.1 Simulation and Tracking Functions

The simulation and tracking infrastructure was provided as part of the assignment. These functions model the physical tracking system, marker-body manufacturing uncertainty, tracker measurement noise, pointer placement, and calibration sensor measurements.

- `define_marker_body(name, nominal_positions)`: Creates a MarkerBody object from the specified nominal marker coordinates. Manufacturing uncertainty is introduced internally.
- `place_marker_body(body, F_approx, precision)`: Places a marker body at an approximate Frame and generates its actual placement using the simulation's uncertainty model. The precision argument distinguishes coarse placement from precise positioning.
- `define_pointer(name, nominal_tip_position, marker_body)`: Creates a Pointer object associated with a marker body and an approximate tip position.
- `create_tracker(name, F_approx)`: Initializes the Tracker object using the approximate camera Frame.
- `sample_marker_bodies(tracker, placed_bodies)`: Simulates one simultaneous tracker observation of the specified marker bodies. Returns estimated body Frames and observed marker coordinates.
- `hand_guide_pointer(pointer, pointer_placed, cal_placed, R_desired)`: Simulates hand-guiding the pointer near the calibration object's origin at the specified orientation.
- `sense_tip(pointer, hand_guide_result)`: Measures the pointer tip position relative to the calibrator. Returns a valid position if the tip is within the sensor's 5 mm range; otherwise, returns None.
- `Tracker.read_marker_body()`: Simulates marker measurements in tracker coordinates while incorporating tracker pose perturbations and marker detection noise.
- `Tracker.procrustes_solver()`: Uses Arun's SVD-based rigid point-set registration to estimate the rotation and translation between nominal marker positions and observed marker positions.
- `MarkerBody.set_marker_positions()`: Updates the nominal marker geometry using refined marker coordinates obtained from repeated observations.

The `sample_marker_bodies()` function internally invokes the tracker's marker-reading and Procrustes registration methods. Consequently, the rigid registration procedure is provided by the simulation framework rather than implemented independently.

4.1.2 Designed Functions

The student-implemented functions perform marker geometry refinement, calibration orientation selection, least-squares tip estimation, and validation of the calibrated pointer.

- `refine_marker_body(trk, placed, n_readings=30, passes=3)`:

  Performs three refinement passes, each containing 30 tracker observations. For each observation, measured marker positions are transformed into the marker body's local coordinate system. The resulting coordinates are averaged to update the nominal marker positions.

  Input: Tracker, placed marker body, number of readings, number of refinement passes.

  Output: Updated nominal marker positions stored in the MarkerBody object.

- `rotation_distance_deg(R1, R2)`:

  Computes the angular separation between two rotations using their relative rotation matrix and the matrix trace. The function is used to reject calibration orientations separated by less than 25 degrees.

  Input: Two rotation matrices.

  Output: Relative rotation angle in degrees.

- `compute_pivot_calibration( F_ptr_list, F_cal_list, sensed_tip_list)`:

  Constructs a linear least-squares system using simultaneous pointer and calibrator Frame observations and calibration sensor measurements.

  The stacked system is solved using NumPy least squares. Residuals are subsequently used to estimate the covariance of the calibrated tip position.

  Input: Pointer Frames, calibrator Frames, and sensed tip positions.

  Output: Calibrated pointer tip position and estimated covariance matrix.

- `xyz(v)`:

  Converts a three-dimensional vector into a flattened NumPy array for use in the 3D visualization routines.

  Input: Three-dimensional vector.

  Output: NumPy array containing the vector coordinates.

Validation and Statistical Analysis The test-pose validation and statistical analysis procedures were implemented directly in the Jupyter notebook rather than as separate functions.

The validation procedure generates a set of candidate test poses and randomly selects 50 unique poses. For each pose, the pointer is positioned using the simulation framework, and exactly one simultaneous tracker observation is collected.

The calibrated tip position is transformed into the nearby reference body's coordinate system and compared against the simulator's ground-truth tip position.

The resulting errors are used to calculate the mean error, maximum error, root mean square error, 95th-percentile error, and the number of test poses satisfying the 2 mm requirement.

Ground-truth information is used exclusively during validation and is not used in the calibration calculations.

4.2 Program Organization

The program is organized into a Jupyter notebook that implements the calibration workflow, supporting source modules that provide simulation and geometric operations, and visualization utilities.

- `assignment1_vinahmed.ipynb`:

  Serves as the main executable program. It contains the marker-body and workspace design parameters, student-implemented functions, calibration data collection, least-squares estimation, test-pose validation, and statistical analysis.

- `src/`:

  Contains the supporting Python packages used by the notebook, including the provided tracking and simulation infrastructure. These modules implement the underlying geometric operations, simulated measurements, and uncertainty models.

- `simulation/`:

  Provides functions for defining and placing marker bodies, initializing the tracker and pointer, collecting simulated tracker measurements, hand-guiding the pointer, and reading the calibration sensor.

  The simulation framework also models manufacturing errors, placement uncertainty, and measurement noise.

- `data_types/`:

  Provides the mathematical representations used by the program, including three-dimensional vectors, rotation matrices, rigid-body Frames, and covariance-related types.

- `functions/`:

  Contains the visualization utility imported by the notebook. The `plot_workspace()` function is used to visualize the tracker, marker bodies, and workspace geometry.

The program begins by initializing the pointer, calibration object, and tracker. The nominal marker geometries are then refined using repeated observations.

Calibration measurements are collected at diverse pointer orientations and passed to the least-squares calibration function.

The estimated pointer tip position is subsequently evaluated over 50 test poses using an additional nearby reference marker body.

Finally, the program computes validation statistics and produces visualizations of the workspace and tip-position errors.

4.3 Key Design Parameters

Parameter	Selected Value
Pointer markers	$(0,0,0)$, $(50,0,0)$, $(0,80,0)$, $(0,0,60)$ mm
Approximate pointer tip	$(0,0,-150)$ mm
Calibrator markers	$(400,-200,0)$, $(400,200,0)$, $(-400,-200,0)$, $(-400,200,100)$ mm
Tracker Frame	$F_{tracker} = (I, (-1100, 300, 300))$
Calibrator Frame	$F_{cal} = (I, (650, -200, 300))$
Marker refinement	3 passes, 30 readings per pass
Calibration observations	30 valid observations
Candidate orientations	XYZ angles between $-70°$ and $70°$
Minimum orientation separation	$25°$
Reference-body markers	$(0,0,0)$, $(400,0,0)$, $(0,500,0)$, $(0,0,600)$ mm
Reference-body Frame	$F_{near} = (I, (600, 300, 300))$
Test rotation angles	$\{0, \pm\pi/6, \pm\pi/3, \pm\pi/2\}$
Test translation offsets	$\{0, \pm100, \pm200\}$ mm about $(300, 300, 300)$ mm
Validation poses	50 randomly selected unique poses
Observations per test pose	1 simultaneous tracker observation
Accuracy requirement	≤ 2 mm

Table 1: Key design parameters for the pivot calibration simulation.

5 Validation Approach

5.1 Unit Tests

5.1.1 Frame Transformations and Rigid Registration Tests

Frame transformations and rigid registration are fundamental operations in the calibration pipeline. Errors in transformation order, frame inversion, or rigid-body registration may propagate to subsequent calibration steps and affect the estimated tool-tip position. To verify these operations independently of the complete simulation, nine synthetic unit tests (UT-01–UT-09) were conducted, covering coordinate-frame consistency, rigid registration accuracy, and invalid-input handling.

Frame Transformation Tests (UT-01–UT-03). The first three tests verified the mathematical consistency of coordinate transformations using predefined rotations, translations, and point coordinates. UT-01 tested frame inversion by transforming a known point and subsequently applying the inverse transformation. UT-02 verified frame composition by comparing sequential and composed transformations against an analytically determined position.

UT-03 specifically examined the transformation chain between the tracker, pointer, and calibration-object coordinate frames. The pointer-tip position expressed in the calibration frame was computed as

$$\mathbf{p}_{\text{sensed}} = F_{\text{cal,ptr}}\,\mathbf{p}_{\text{tip}} = F_{\text{cal}}^{-1} F_{\text{ptr}}\,\mathbf{p}_{\text{tip}}, \tag{78}$$

Here, $F_{\text{ptr},j}$ and $F_{\text{cal},j}$ denote the pointer and calibration-object poses relative to the tracking camera in observation j. They should be distinguished from $F_{\text{cal}}^{\text{approx}}$, which denotes the approximate calibration-object placement relative to the workspace. The relative transformation from the pointer frame to the calibration frame is given by $F_{\text{cal,ptr},j} = F_{\text{cal},j}^{-1} F_{\text{ptr},j}$.

The computed position was compared with an independently determined analytical result. For UT-01–UT-03, the positional error was evaluated using

$$e_p = \|\mathbf{p}_{\text{computed}} - \mathbf{p}_{\text{expected}}\|_2 \,, \tag{79}$$

with a numerical tolerance of 10^{-8} mm.

Rigid Registration Tests (UT-04–UT-07). The registration tests evaluated the course-provided SVD-based rigid registration solver using synthetic marker correspondences. Given known marker positions $\mathbf{a}_k$ and a predefined rigid transformation $(R_{\text{true}}, \mathbf{t}_{\text{true}})$, observations were generated according to

$$\mathbf{b}_k = R_{\text{true}}\mathbf{a}_k + \mathbf{t}_{\text{true}} + \boldsymbol{\epsilon}_k, \tag{80}$$

where $\boldsymbol{\epsilon}_k$ denotes the simulated marker measurement error.

UT-04 and UT-05 evaluated exact transformation recovery using three noncollinear and five noncoplanar marker correspondences, respectively, with $\boldsymbol{\epsilon}_k = \mathbf{0}$. Both tests required translation error below 10^{-6} mm and rotation error below 10^{-4} degrees.

To evaluate numerical robustness, UT-06 and UT-07 introduced independent, zero-mean Gaussian noise with standard deviations of $\sigma = 0.05$ mm and $\sigma = 0.20$ mm, respectively. The estimated rigid transformations were evaluated using translation and rotation errors:

$$e_t = \left\|\hat{\mathbf{t}} - \mathbf{t}_{\text{true}}\right\|_2 \tag{81}$$

$$e_R = \cos^{-1}\left(\frac{\text{tr}\left(\hat{R}R_{\text{true}}^T\right) - 1}{2}\right) \tag{82}$$

The rotation error was converted to degrees for evaluation. The acceptance thresholds for UT-06 were 0.5 mm and 0.5 degrees, while those for UT-07 were 1.0 mm and 1.0 degrees. A fixed random seed was used to ensure reproducibility of the synthetic noisy observations.

Invalid-Input Tests (UT-08–UT-09). Finally, two tests examined the solver's handling of invalid marker correspondences. UT-08 provided only two marker pairs, while UT-09 supplied mismatched numbers of nominal and observed markers. Both tests required the solver to reject the input by raising a `ValueError`.

All nine unit tests satisfied their predefined acceptance criteria. These results verify the correctness of the tested frame-transformation operations and the numerical behavior of rigid registration under the specified synthetic conditions. The detailed numerical results are presented in Section 6.2.

5.1.2 Marker Geometry Refinement Tests

Marker geometry refinement is used to improve the estimated local positions of markers through repeated tracker observations. Since inaccuracies in marker geometry may propagate to rigid registration and subsequent tip calibration, five synthetic unit tests (UT-10–UT-14) were conducted to verify the coordinate transformation, observation averaging, iterative updating, and data consistency of the implemented `refine_marker_body()` function.

The refinement algorithm transforms each observed marker position from the tracker frame into the corresponding marker-body frame. For marker k in observation j, the estimated local position is

$$\hat{\mathbf{a}}_{k,j} = \hat{F}_j^{-1}\mathbf{b}_{k,j}, \tag{83}$$

where $\mathbf{b}_{k,j}$ is the observed marker position and $\hat{F}_j$ is the estimated marker-body pose relative to the tracker. The local marker positions are then averaged over N observations:

$$\bar{\mathbf{a}}_k = \frac{1}{N}\sum_{j=1}^{N}\hat{\mathbf{a}}_{k,j}. \tag{84}$$

The resulting estimates are used to update the marker-body geometry, and the procedure can be repeated over multiple refinement passes.

To isolate the refinement logic from stochastic tracker and registration errors, predefined marker observations and known rigid transformations were provided through mocked tracker measurements. This allowed the expected local marker positions and update behavior to be determined independently.

Exact Geometry Recovery (UT-10). Four synthetic marker positions were transformed using three known rigid transformations without measurement noise. The refinement function was expected to recover the original local marker coordinates after applying the inverse transformations and averaging the observations. The maximum Euclidean marker-position error was required to remain below 10^{-8} mm.

Symmetric Noise Averaging (UT-11). Three synthetic observations were generated with positive, negative, and zero marker-position perturbations, respectively. Since the perturbations were constructed to cancel exactly, their expected average corresponded to the original marker geometry. This test verified the implementation of observation averaging using a numerical tolerance of 10^{-8} mm.

Multi-Pass Geometry Updates (UT-12). Two refinement passes were performed with two observations per pass. Controlled marker-position offsets were introduced to produce known mean coordinates for each pass. The test verified that the geometry was updated twice, that four tracker readings were requested, and that both updates matched the corresponding analytical averages.

Marker Correspondence Preservation (UT-13). Distinct perturbations were assigned to four markers to verify that refinement preserved the number and ordering of marker correspondences. The updated marker positions were compared with their individually prescribed coordinates using a tolerance of 10^{-8} mm.

Zero-Pass Behavior (UT-14). The refinement function was executed with zero refinement passes. The test verified that no tracker readings or geometry updates occurred and that the original marker positions remained unchanged.

All five tests passed their predefined acceptance criteria. The results confirm the correctness of the implemented coordinate transformations, averaging operations, iterative updates, and marker correspondence handling under the tested synthetic conditions. Since the tests used known marker-body poses, they do not independently establish the accuracy of geometry refinement under realistic tracker uncertainty. This aspect is evaluated separately in Section **??**. The numerical unit-test results are summarized in Section 6.2.

5.1.3 Calibration Pose Generation and Rotation Diversity Tests

Calibration pose generation is an essential component of the observation collection procedure. To obtain diverse pointer orientations, the implemented algorithm randomly generates candidate rotations and rejects those that are insufficiently different from previously accepted orientations. Six unit tests (UT-15–UT-20) were designed to verify the rotation distance calculation, sampling constraints, orientation diversity, and reproducibility of this procedure.

Rotation Distance Tests (UT-15–UT-17). The angular distance between two rotations, R_1 and R_2, was computed using the relative rotation matrix:

$$d(R_1, R_2) = \cos^{-1}\left(\frac{\mathrm{tr}(R_1^T R_2) - 1}{2}\right), \tag{85}$$

where the resulting angle is converted to degrees. To prevent numerical errors in the inverse cosine calculation, its argument is constrained to the interval $[-1, 1]$.

Three tests verified the correctness of the implemented `rotation_distance_deg()` function. UT-15 evaluated the angular distance between a rotation and itself, with an expected value of $0°$. UT-16 compared the identity rotation with a known $90°$ rotation about the X-axis, while UT-17 verified the symmetry property

$$d(R_1, R_2) = d(R_2, R_1). \tag{86}$$

The acceptance tolerances were 10^{-4} degrees for UT-15 and UT-16, and 10^{-6} degrees for UT-17.

Random Pose Generation Test (UT-18). The implemented `sample_diverse_rotation()` function generates candidate orientations using XYZ Euler angles independently sampled from the range

$$\theta_x, \theta_y, \theta_z \in [-70°, 70°]. \tag{87}$$

Thirty candidate orientations were generated using a fixed random seed. UT-18 verified that every sampled Euler angle remained within the specified bounds.

Rotation Diversity Test (UT-19). To avoid repeatedly collecting observations at similar orientations, each candidate rotation is compared against all previously accepted rotations. A candidate is accepted only if

$$d(R_i, R_j) \geq 25^\circ, \qquad \forall\, i \neq j, \tag{88}$$

for all accepted rotation pairs.

UT-19 generated 30 accepted orientations and calculated the minimum pairwise angular distance. The test also verified that a candidate rotation separated by 10° from an existing rotation was rejected, whereas a candidate separated by 40° was accepted. The test passed if all accepted rotations satisfied the minimum 25° separation requirement and both explicit filtering checks behaved as expected.

Reproducibility Test (UT-20). Finally, UT-20 evaluated the reproducibility of the random pose generation procedure. Two sequences of 20 orientations were generated using the same random seed, while a third sequence used a different seed. The test required identical sequences for the same seed and a different sequence for the alternative seed.

All six unit tests passed their predefined acceptance criteria. The results verify the correctness of the implemented rotation-distance calculation, the specified Euler-angle sampling bounds, the minimum angular separation constraint, and deterministic pose generation under fixed random seeds. Detailed numerical results are summarized in Section 6.2.

5.1.4 Tip Calibration Solver Tests

The tip calibration solver is a core computational component of the proposed system. It estimates the pointer-tip position in the pointer coordinate frame by combining the tracker-observed pointer and calibrator poses with measurements from the calibration sensor. Eight synthetic unit tests (UT-21–UT-28) were designed to verify the correctness of the least-squares solution, covariance estimation, observation filtering, and numerical behavior under controlled measurement errors.

For each valid calibration observation j, the unknown pointer-tip position satisfies

$$R_{\text{ptr},j}\mathbf{p}_{\text{tip}} = R_{\text{cal},j}\mathbf{p}_{\text{sensed},j} + \mathbf{t}_{\text{cal},j} - \mathbf{t}_{\text{ptr},j}, \tag{89}$$

where $R_{\text{ptr},j}$ and $\mathbf{t}_{\text{ptr},j}$ describe the pointer pose relative to the tracker, while $R_{\text{cal},j}$ and $\mathbf{t}_{\text{cal},j}$ describe the calibrator pose relative to the tracker. The sensor measurement $\mathbf{p}_{\text{sensed},j}$ is expressed in the calibrator frame.

The equations from all valid observations are stacked into a linear system,

$$A\mathbf{x} = \mathbf{b}, \qquad \hat{\mathbf{x}} = \arg\min_{\mathbf{x}} \|A\mathbf{x} - \mathbf{b}\|_2^2, \tag{90}$$

where $\mathbf{x} = \mathbf{p}_{\text{tip}}$ is the three-dimensional unknown tip position.

Exact Synthetic Solution Tests (UT-21–UT-22). UT-21 evaluated the solver using three predefined tip positions and observation counts of $N = 3$, 10, and 30. Synthetic sensor readings were generated from known rigid transformations, ensuring that the true tip position was available independently of the solver. Each reconstructed tip position was required to agree with the known value within 10^{-8} mm.

UT-22 examined the minimum-data condition using a single valid observation. Since the coefficient matrix for one observation is a 3×3 rotation matrix, the resulting system has full column rank and is mathematically solvable. However, a single observation provides no residual degrees of freedom for estimating measurement variance. Therefore, the test required an accurate tip estimate and an unavailable covariance estimate represented by `NaN` values.

Least-Squares Accuracy and Covariance Tests (UT-23–UT-24). UT-23 introduced predefined measurement perturbations into four synthetic observations with identity transformations. Under these conditions, the least-squares solution has a known analytical form, equal to the true tip position plus the mean measurement perturbation. The computed solution was compared with this analytical result using a tolerance of 10^{-8} mm.

UT-24 verified the covariance calculation. The residual-based covariance estimate was computed as

$$\hat{\Sigma}_{\text{tip}} = \hat{\sigma}^2 (A^T A)^+, \qquad \hat{\sigma}^2 = \frac{\|A\hat{\mathbf{x}} - \mathbf{b}\|_2^2}{3N_{\text{valid}} - \text{rank}(A)}, \tag{91}$$

where $(\cdot)^+$ denotes the Moore–Penrose pseudoinverse and N_{valid} is the number of valid observations. This estimate assumes independent, isotropic residual errors in the stacked linear system.

For four identity-rotation observations, $A^T A = 4I$, allowing the expected covariance to be calculated analytically. The maximum difference between the estimated and expected covariance matrices was required to be below 10^{-10} mm^2.

Invalid-Input Handling Tests (UT-25–UT-27). UT-25 introduced two invalid sensor measurements, represented by `None`, into five synthetic observations. The solver was required to discard invalid measurements while preserving the correspondence between the remaining tracker poses and sensor readings.

UT-26 verified that the solver rejected a dataset containing no valid sensor observations. UT-27 examined mismatched input-list lengths. Both tests required a `ValueError` to be raised.

Measurement Error Scaling Test (UT-28). Finally, UT-28 evaluated the numerical scaling properties of the least-squares solution and its covariance estimate. The predefined measurement perturbations were multiplied by a factor of two while keeping the transformation matrices unchanged. For this controlled linear system, the expected estimation bias doubles, whereas the residual-based covariance scales by a factor of four:

$$\Delta\mathbf{p}_{\text{tip}}^{(2)} = 2\Delta\mathbf{p}_{\text{tip}}^{(1)}, \qquad \hat{\Sigma}_{\text{tip}}^{(2)} = 4\hat{\Sigma}_{\text{tip}}^{(1)}. \tag{92}$$

The numerical differences from these expected relationships were required to remain below 10^{-10} in the corresponding units.

All eight tests passed their predefined acceptance criteria. These results confirm the mathematical correctness and numerical consistency of the implemented tip calibration solver under the specified synthetic conditions. The tests do not independently establish the final positioning accuracy under stochastic tracker and sensor errors, which is evaluated separately in the subsequent validation procedures. Detailed numerical results are provided in Section 6.2.

5.1.5 Tip Position Prediction Tests

Tip position prediction is the final computational component of the calibration pipeline. It transforms the calibrated pointer-tip position into the coordinate frame of the nearby reference marker body using the tracker-observed rigid-body poses. Seven synthetic unit tests (UT-29–UT-35) were conducted to verify the correctness of the transformation chain, coordinate-frame consistency, calibration error propagation, and positioning accuracy evaluation.

The predicted tip position relative to the nearby reference body is computed as

$$\mathbf{p}_{\text{predicted}} = F_{\text{near}}^{-1} F_{\text{ptr}} \mathbf{p}_{\text{tip}}, \tag{93}$$

where F_{ptr} and F_{near} represent the pointer and nearby reference body poses relative to the tracker, respectively, and $\mathbf{p}_{\text{tip}}$ is the calibrated tip position in the pointer frame.

Coordinate Transformation Tests (UT-29–UT-31). UT-29 verified tip position prediction under pure translation using predefined pointer and reference-body poses. The analytically determined target position was $(50, 100, 0)$ mm, and the numerical result was required to agree within 10^{-8} mm.

UT-30 examined the transformation chain involving a rotated reference frame. A 90° rotation about the Z-axis was applied to the reference body, together with predefined translations. The expected tip position in the reference frame was analytically determined as $(10, -20, 10)$ mm. This test was designed to detect incorrect transformation order and improper application of inverse rotations.

UT-31 considered the special case where the pointer and reference body shared an identical pose. Since the two transformations cancel, the predicted tip position should equal the original local tip coordinates. The numerical tolerance was 10^{-8} mm.

Tracker-Frame Invariance Test (UT-32). UT-32 verified that the predicted relative tip position is independent of a common change in the tracker coordinate representation. An arbitrary rigid transformation G was applied to both the pointer and reference-body poses. The prediction should remain unchanged because

$$(GF_{\text{near}})^{-1}(GF_{\text{ptr}})\mathbf{p}_{\text{tip}} = F_{\text{near}}^{-1} F_{\text{ptr}} \mathbf{p}_{\text{tip}}. \tag{94}$$

The predicted positions before and after this transformation were compared using a tolerance of 10^{-8} mm.

Calibration Error Propagation Test (UT-33). UT-33 investigated the propagation of a known tip calibration error through the prediction transformation. A predefined offset of $(0.6, -0.8, 0)$ mm was added to the true local tip position, producing an error magnitude of 1 mm.

With exact rigid-body poses, the predicted tip-position difference is

$$\Delta\mathbf{p}_{\text{predicted}} = R_{\text{near}}^{T} R_{\text{ptr}} \Delta\mathbf{p}_{\text{tip}}. \tag{95}$$

Since rigid rotations preserve Euclidean distance,

$$\|\Delta\mathbf{p}_{\text{predicted}}\|_2 = \|\Delta\mathbf{p}_{\text{tip}}\|_2 = 1 \text{ mm}. \tag{96}$$

The test required the computed positioning error to match the analytical value within 10^{-8} mm.

Positioning Error and Threshold Tests (UT-34–UT-35). UT-34 verified the Euclidean positioning error calculation:

$$e_{\text{tip}} = \|\mathbf{p}_{\text{predicted}} - \mathbf{p}_{\text{actual}}\|_2 . \tag{97}$$

Two predefined point pairs were used, with expected distances of 13 mm and 0 mm, respectively.

Finally, UT-35 examined the boundary behavior of the 2 mm positioning accuracy criterion. Errors of 1.9999 mm, 2.0000 mm, and 2.0001 mm were tested to verify that the implemented condition accepts errors less than or equal to 2 mm and rejects errors exceeding the threshold.

All seven tests passed their predefined acceptance criteria. These results verify the mathematical consistency of the implemented tip prediction and error evaluation functions under controlled synthetic conditions. The final positioning accuracy under simulated tracker uncertainty is evaluated separately through the end-to-end validation procedure. Detailed unit-test results are presented in Section 6.2.

5.2 Calibration Validation

In addition to unit testing, calibration-specific validation was performed using measurements generated by the simulation environment. Unlike the synthetic unit tests, these experiments incorporated the simulator's measurement uncertainty and evaluated the calibration pipeline using observations not included in the corresponding estimation procedures.

Two validation experiments were conducted: marker geometry refinement verification and held-out calibration sensor validation.

5.2.1 Marker Geometry Refinement Verification

The marker geometry refinement procedure was evaluated to determine whether the estimated marker configuration became more consistent with independent tracker observations.

For both the pointer and calibrator marker bodies, the original nominal marker positions were recorded before refinement. The refinement procedure then estimated updated local marker positions by transforming repeated tracker observations into the corresponding marker-body coordinate frame and averaging the results over multiple observations.

To evaluate the refinement independently of changes in coordinate-frame representation, pairwise distances between marker spheres were used instead of directly comparing their local coordinates. For marker positions $\mathbf{a}_i$ and $\mathbf{a}_j$, the modeled distance is

$$d_{ij}^{\text{model}} = \|\mathbf{a}_i - \mathbf{a}_j\|_2 . \tag{98}$$

This distance is invariant under rigid transformations and therefore provides a coordinate-frame-independent description of the marker-body geometry.

After refinement, 20 additional tracker observations were collected for each marker body. For each observation, the corresponding distances between the observed marker positions were calculated.

The same validation observations were compared against both the original and refined marker geometries.

The marker-distance root mean square error was defined as

$$\mathrm{RMSE}_d = \sqrt{\frac{1}{NM}\sum_{n=1}^{N}\sum_{i<j}\left(d_{ij}^{\mathrm{model}} - d_{ij,n}^{\mathrm{observed}}\right)^2}, \tag{99}$$

where $N = 20$ is the number of validation observations and M is the number of unique marker pairs. For the four-marker configurations used in this assignment, $M = 6$.

The relative reduction in marker-distance RMSE was calculated as

$$\mathrm{Reduction} = \frac{\mathrm{RMSE}_{\mathrm{before}} - \mathrm{RMSE}_{\mathrm{after}}}{\mathrm{RMSE}_{\mathrm{before}}} \times 100\%. \tag{100}$$

This experiment assesses agreement between the marker geometry model and newly acquired observations without requiring access to the simulator's hidden true marker positions. The numerical results are reported in Section 6.3.

5.2.2 Held-Out Sensor Validation

The calibrated pointer-tip position was further evaluated using a held-out dataset consisting of new calibration observations that were not used to compute the least-squares estimate.

First, residuals were evaluated for the 30 valid observations used in the calibration solver. These training residuals characterize the agreement between the fitted calibration model and the observations used during estimation.

Next, 10 additional valid observations were collected using newly generated pointer orientations. For each observation j, the estimated tip position in the calibrator coordinate frame was computed as

$$\hat{\mathbf{p}}_{\mathrm{sensed},j} = F_{\mathrm{cal},j}^{-1} F_{\mathrm{ptr},j} \hat{\mathbf{p}}_{\mathrm{tip}}, \tag{101}$$

where $F_{\mathrm{ptr},j}$ and $F_{\mathrm{cal},j}$ are the simultaneously observed pointer and calibrator poses, and $\hat{\mathbf{p}}_{\mathrm{tip}}$ is the previously estimated pointer-tip position.

The predicted tip position was then compared with the corresponding calibration sensor measurement:

$$e_j^{\mathrm{sensor}} = \left\| \hat{\mathbf{p}}_{\mathrm{sensed},j} - \mathbf{p}_{\mathrm{sensed},j} \right\|_2 . \tag{102}$$

Sensor readings that were invalid were excluded, and additional observations would be collected until the required number of valid held-out measurements was obtained.

The following evaluation metrics were calculated:

$$\bar{e}_{\text{sensor}} = \frac{1}{N}\sum_{j=1}^{N} e_j^{\text{sensor}}, \tag{103}$$

$$\text{RMSE}_{\text{sensor}} = \sqrt{\frac{1}{N}\sum_{j=1}^{N}\left(e_j^{\text{sensor}}\right)^2}, \tag{104}$$

$$e_{\text{sensor,max}} = \max_{1\le j\le N} e_j^{\text{sensor}}. \tag{105}$$

The mean residuals from the calibration and held-out datasets were compared to assess whether the estimated tip position remained consistent with new sensor observations.

Unlike the final workspace positioning evaluation, this experiment measures agreement with the calibration sensor rather than the positioning error relative to the true pointer-tip location. Its purpose is to verify the consistency of the estimated calibration parameters on previously unseen measurements. The results are presented in Section 6.3.

5.3 End-to-End Validation

End-to-end validation was performed to evaluate the positioning accuracy of the complete calibration pipeline over the prescribed test workspace. Unlike the preceding unit tests and calibration-specific validation experiments, this procedure considers the combined effects of pointer-tip calibration uncertainty, marker geometry errors, and tracker pose estimation errors.

Following Step 8 of the assignment, the calibrated pointer was placed at a set of nominal test poses, and its tip position was estimated using a single tracker observation at each pose. The predicted positions were subsequently compared with simulator ground-truth values, which were used exclusively for performance evaluation.

5.3.1 Test Pose Generation and Single-Observation Protocol

The nominal test poses were generated according to the position and orientation specifications defined in the assignment. Each test pose was represented by a rigid transformation

$$F_{t,i} = \begin{bmatrix} R(\theta_x, \theta_y, \theta_z) & \mathbf{t}_i \\ \mathbf{0}^T & 1 \end{bmatrix}, \tag{106}$$

where the orientation was constructed from XYZ Euler angles selected from

$$\theta_x, \theta_y, \theta_z \in \left\{0, \pm\frac{\pi}{6}, \pm\frac{\pi}{3}, \pm\frac{\pi}{2}\right\}, \tag{107}$$

and the nominal translation was defined by

$$\mathbf{t}_i = \begin{bmatrix} 300 + \delta_x \\ 300 + \delta_y \\ 300 + \delta_z \end{bmatrix} \text{mm}, \quad \delta_x, \delta_y, \delta_z \in \{0, \pm 100, \pm 200\}. \tag{108}$$

The complete nominal grid consists of $7^3 \times 5^3 = 42,875$ possible pose combinations. For the extended end-to-end evaluation, 10,000 distinct nominal test poses were randomly selected without replacement from this grid.

A fixed NumPy random seed of 42 was used for est-pose selection, ensuring that the same nominal pose set could be reproduced across independent simulation runs. Other stochastic processes were not intentionally fixed to this pose-selection seed, allowing the complete calibration and positioning pipeline to exhibit run-to-run variation.

For each selected nominal pose $F_{t,i}$, the simulator placed the pointer at an actual pose incorporating a small positioning perturbation. This represented the uncertainty associated with positioning the pointer in a physical system.

An additional marker body was placed near, but outside, the prescribed test volume and was used as a local reference frame. Its marker geometry was refined before the final validation procedure.

For each test pose, exactly one simultaneous tracker observation was acquired to obtain the estimated pointer and nearby reference-body poses, denoted by $F^{\text{obs}}_{\text{ptr},i}$ and $F^{\text{obs}}_{\text{near},i}$, respectively. No observation averaging or repeated tracker measurements were performed at individual test poses.

The predicted pointer-tip position in the nearby reference-body coordinate frame was calculated using the previously calibrated tip position $\hat{\mathbf{p}}_{\text{tip}}$:

$$\hat{\mathbf{p}}^{\text{near}}_{\text{tip},i} = \left(F^{\text{obs}}_{\text{near},i}\right)^{-1} F^{\text{obs}}_{\text{ptr},i} \hat{\mathbf{p}}_{\text{tip}}. \tag{109}$$

This prediction relies only on the estimated tracker poses and calibrated pointer-tip geometry. Hidden simulator ground-truth parameters were not used in the prediction process.

5.3.2 Error Metrics and Evaluation Criteria

The simulator provides access to the actual rigid-body placements and pointer-tip geometry during debugging and validation. These hidden quantities were used only to calculate the true tip position for error evaluation.

For each test pose i, the actual tip position in the nearby reference-body frame was computed as

$$\mathbf{p}^{\text{true,near}}_{\text{tip},i} = \left(F^{\text{true}}_{\text{near}}\right)^{-1} F^{\text{true}}_{\text{ptr},i} \mathbf{p}^{\text{true}}_{\text{tip}}, \tag{110}$$

where $F^{\text{true}}_{\text{ptr},i}$ and $F^{\text{true}}_{\text{near}}$ represent the actual pointer and reference-body placements, respectively.

Because marker geometry refinement updates the local marker coordinate model, the refined marker-body coordinate frame may differ from the original physical coordinate frame used by the simulator.

Consequently, the predicted tip coordinates and the ground-truth tip coordinates must be expressed in a consistent reference frame before calculating their Euclidean distance.

For validation purposes, a rigid transformation $F_{\text{align,near}}$ was determined by registering the refined near-reference marker geometry to the simulator's actual marker geometry.

This transformation maps points from the refined near-reference coordinate system to the original physical near-reference coordinate system.

The coordinate-consistent predicted tip position was calculated as

$$\mathbf{p}_{\text{tip},i}^{\text{aligned}} = F_{\text{align,near}} \hat{\mathbf{p}}_{\text{tip},i}^{\text{near}}. \tag{111}$$

The corrected Euclidean positioning error was therefore defined as

$$e_i = \left\| \mathbf{p}_{\text{tip},i}^{\text{aligned}} - \mathbf{p}_{\text{tip},i}^{\text{true,near}} \right\|_2. \tag{112}$$

Equivalently, substituting the pointer-tip prediction equation gives

$$\begin{aligned} e_i = \Big\| & F_{\text{align,near}} \left(F_{\text{near},i}^{\text{obs}} \right)^{-1} F_{\text{ptr},i}^{\text{obs}} \hat{\mathbf{p}}_{\text{tip}} \\ & - \left(F_{\text{near}}^{\text{true}} \right)^{-1} F_{\text{ptr},i}^{\text{true}} \mathbf{p}_{\text{tip}}^{\text{true}} \Big\|_2. \end{aligned} \tag{113}$$

The alignment transformation was derived using simulator ground truth exclusively for retrospective validation. It was not used during pointer-tip calibration or operational tip-position prediction.

This distinction ensures that the evaluation compares geometrically equivalent points without introducing ground-truth information into the estimation pipeline.

Because both tip positions are expressed in the same reference frame, their Euclidean distance provides a coordinate-frame-independent measure of the positioning error.

To characterize the system performance over the $N = 10,000$ test poses in each extended run, several complementary error metrics were computed.

Mean Positioning Error. The average magnitude of the positioning error was calculated as

$$\bar{e} = \frac{1}{N} \sum_{i=1}^{N} e_i. \tag{114}$$

Root Mean Square Error. The root mean square error (RMSE) was used to characterize the overall error magnitude while assigning greater weight to larger errors:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} e_i^2}. \tag{115}$$

Maximum Error and 95th-Percentile Error. The maximum error was defined as

$$e_{\max} = \max_{1 \le i \le N} e_i. \tag{116}$$

The empirical 95th-percentile error, denoted by e_{95}, was also calculated to describe the upper range of the observed error distribution.

Two-Millimeter Accuracy Criterion. Each test pose was classified according to whether its positioning error satisfied the 2 mm requirement:

$$\text{Pass}_i = \begin{cases} 1, & e_i \le 2 \text{ mm}, \\ 0, & e_i > 2 \text{ mm}. \end{cases} \tag{117}$$

The empirical passing rate was computed as

$$P_{\text{pass}} = \frac{1}{N} \sum_{i=1}^{N} \text{Pass}_i \times 100\%. \tag{118}$$

In addition to these numerical metrics, the distribution of positioning errors was visualized using a histogram. The histogram displays the percentage of tested poses within each error interval, together with reference lines indicating the 95th-percentile error and the 2 mm accuracy threshold.

These metrics provide complementary descriptions of typical error magnitude, error variability, and the worst observed performance across the selected test poses.

Since each extended run evaluates 10,000 nominal poses rather than the complete grid of 42,875 possible combinations, the resulting statistics represent empirical performance over the sampled configurations.

The corrected coordinate-consistent error metric is used to evaluate physical positioning accuracy, while the original unaligned error measurements are retained separately for diagnostic analysis.

The numerical results and the investigation of coordinate-frame consistency are presented in Section 6.4.

5.4 Repeatability and Sensitivity Testing

Repeatability and sensitivity experiments were conducted to investigate the influence of near-reference marker geometry on the positioning accuracy of the complete calibration system.

During the design optimization process, the calibration algorithm, marker refinement procedure, test-pose generation method, and end-to-end error evaluation protocol were kept unchanged. The near-reference marker geometry was varied as the primary design variable.

Multiple simulation runs were performed for each configuration to account for performance variations caused by stochastic simulation uncertainty.

5.4.1 Repeatability Across Simulation Runs

Repeatability was evaluated by running the complete calibration and validation pipeline under five simulation seed conditions, numbered 42 through 46.

For each run, 50 test poses were evaluated using the same test-pose selection procedure and single-observation protocol described in Section 5.3.1.

For each configuration and simulation run, the number of poses satisfying the 2 mm positioning accuracy requirement was recorded.

The passing rate for run r was defined as

$$P_{\text{pass}}^{(r)} = \frac{N_{\text{pass}}^{(r)}}{50} \times 100\%. \tag{119}$$

The overall passing rate across $K = 5$ runs was calculated as

$$P_{\text{overall}} = \frac{\sum_{r=1}^{K} N_{\text{pass}}^{(r)}}{50K} \times 100\%. \tag{120}$$

Comparing the five runs also allowed the variability of positioning performance to be examined under different stochastic realizations.

The resulting repeatability measurements were used to assess whether a particular marker geometry consistently satisfied the positioning accuracy requirement, rather than relying on a single favorable simulation run.

5.4.2 Sensitivity to Marker Geometry

The sensitivity of positioning performance to near-reference marker spacing was investigated through two stages of controlled design comparison.

The four-marker reference geometry was parameterized as

$$\begin{aligned} \mathbf{a}_1 &= (0, 0, 0)^T, \\ \mathbf{a}_2 &= (L_x, 0, 0)^T, \\ \mathbf{a}_3 &= (0, L_y, 0)^T, \\ \mathbf{a}_4 &= (0, 0, L_z)^T, \end{aligned} \tag{121}$$

where L_x, L_y, and L_z denote the nominal marker spacing along the three coordinate axes.

In the first stage, four symmetric marker configurations were examined:

$$L_x = L_y = L_z \in \{180, 260, 340, 600\} \text{ mm}. \tag{122}$$

The purpose of this comparison was to examine how increasing the marker baseline affected the accuracy of the reference-frame estimation and, consequently, the final pointer-tip positioning error.

In the second stage, the symmetric $(600, 600, 600)$ mm configuration was compared against an asymmetric $(400, 500, 600)$ mm configuration.

The asymmetric design was investigated as a possible compromise between marker spacing, geometric stability, and spatial layout constraints.

For both stages, the same calibration and validation algorithms were retained. Each candidate geometry was evaluated using five simulation seed conditions and 50 test poses per run.

The primary comparison metric was the proportion of evaluated poses satisfying the 2 mm positioning requirement.

Although simulation seeds were controlled in the recorded comparisons, modifying marker geometry could change the subsequent numerical behavior of the simulation. Therefore, the comparisons characterize the observed effect of each geometry under the tested conditions rather than guaranteeing an identical realization of every underlying noise contribution.

The resulting sensitivity and repeatability measurements are presented in Section 6.5.

6 Final Results and Discussion

6.1 Final System Configuration

The final calibration system consists of a tracked pointer, a calibration object, an optical tracker, and an additional nearby reference marker body. The marker geometries and workspace placements were selected to support stable rigid-body registration, pointer-tip calibration, and accurate tip localization over the prescribed test workspace.

Marker-Body Geometry.

Each marker body contains four non-coplanar markers. The designed nominal marker positions, expressed in their respective local coordinate frames, are summarized in Table 2.

Marker	Pointer (mm)	Calibrator (mm)	Near Reference (mm)
1	$(0, 0, 0)$	$(400, -200, 0)$	$(0, 0, 0)$
2	$(50, 0, 0)$	$(400, 200, 0)$	$(400, 0, 0)$
3	$(0, 80, 0)$	$(-400, -200, 0)$	$(0, 500, 0)$
4	$(0, 0, 60)$	$(-400, 200, 100)$	$(0, 0, 600)$

Table 2: Designed nominal marker coordinates of the three marker bodies, expressed in their local coordinate frames.

The pointer marker body uses a relatively compact geometry, with nominal axis-aligned marker separations of 50, 80, and 60 mm. The approximate pointer-tip position in the pointer frame was initialized as

$$\mathbf{p}_{\text{tip,approx}} = (0, 0, -150)^T \text{ mm}. \tag{123}$$

The calibration object employs a larger, non-coplanar marker configuration. The nearby reference

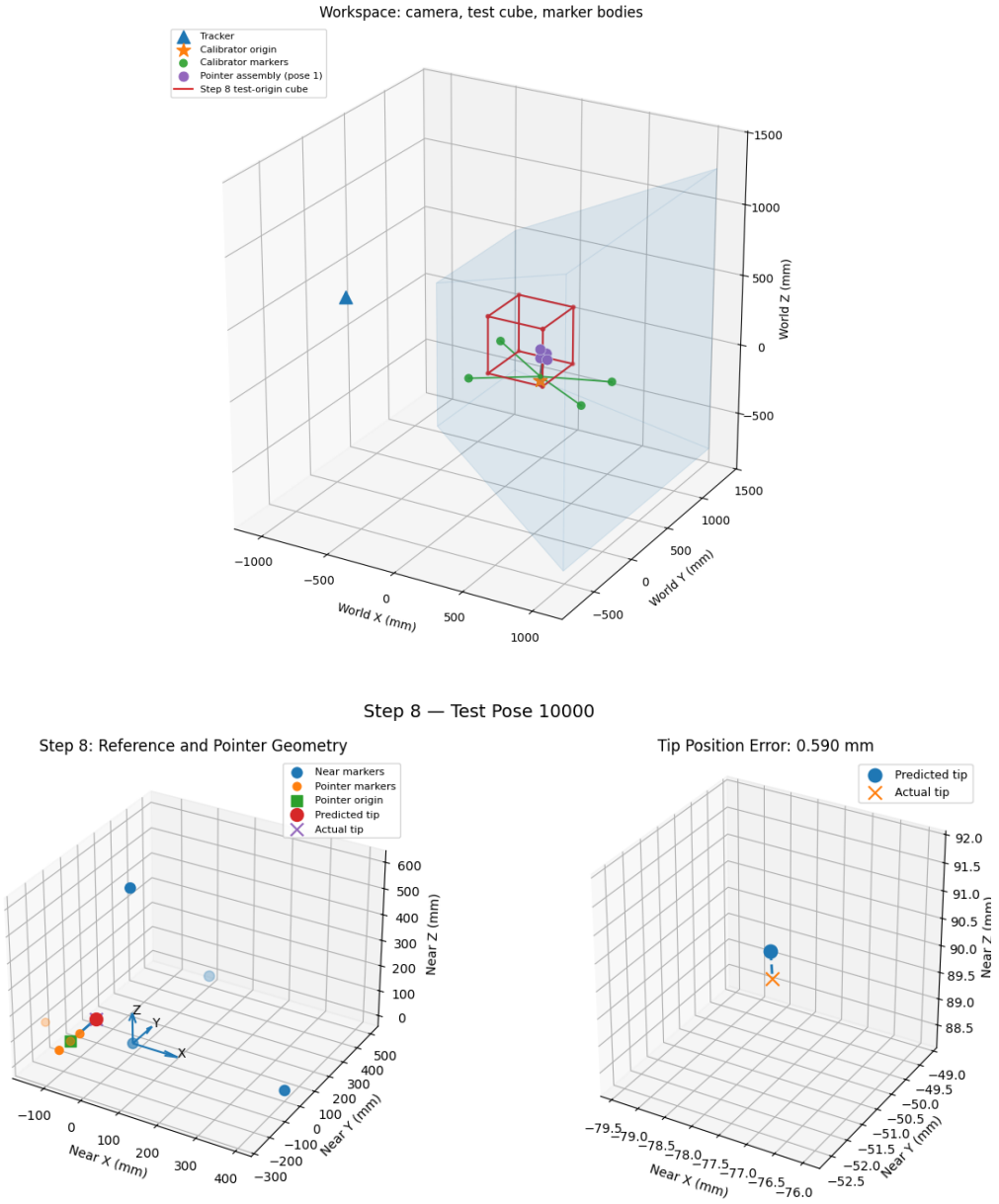


Figure 2: Visualization of the final calibration system. (a) Three-dimensional workspace configuration, including the tracker, calibrator, marker bodies, and prescribed test volume. (b) Representative Step 8 test pose showing the near-reference and pointer marker geometry, together with the predicted and actual tip positions.

marker body uses an asymmetric geometry with axis-aligned marker separations of 400, 500, and 600 mm. This geometry was selected after evaluating the influence of marker spacing on end-to-end positioning performance, as discussed in Section 6.5.

Before calibration and final validation, the marker-body geometries were refined using repeated tracker observations. Therefore, the nominal coordinates in Table 2 describe the initial design rather than the subsequently refined marker coordinates.

Workspace Configuration.

The tracker, calibrator, and nearby reference body were initialized with identity rotations and the approximate translations listed in Table 3.

Component	**Nominal Translation (mm)**	**Rotation**
Tracker	$(-1100, 300, 300)$	Identity
Calibration object	$(650, -200, 300)$	Identity
Nearby reference body	$(600, 300, 300)$	Identity

Table 3: Approximate workspace placements of the stationary system components.

The nearby reference marker body was positioned adjacent to the prescribed test volume and served as the reference coordinate frame for the final pointer-tip positioning evaluation.

All listed placements are nominal configurations. The simulator introduces placement and measurement uncertainties, so the actual simulated rigid-body placements may differ from these values.

Calibration and Validation Parameters.

The primary numerical settings of the final pipeline are summarized in Table 4.

Parameter	**Selected Setting**
Marker bodies	3 (pointer, calibrator, reference)
Markers per body	4
Refinement observations per pass	30
Refinement passes per body	3
Valid tip-calibration observations	30
Calibration orientation sampling	XYZ Euler angles in $[-70°, 70°]$
Minimum accepted rotation separation	25°
Tip calibration method	Linear least squares
Validation poses per extended run	10,000
Tracker observations per test pose	1
Pose-selection random seed	42
Positioning accuracy threshold	2 mm

Table 4: Principal parameters used in the calibration pipeline and extended positioning evaluation.

The calibration procedure estimated the pointer-tip position from simultaneous pointer and calibrator pose measurements together with the calibration sensor readings. The resulting tip estimate was then used without ground-truth correction during the independent positioning evaluations.

Overall, the final configuration combines refined marker geometry, multiple calibration orientations, a least-squares tip estimator, and a nearby reference body to improve positioning performance across the test workspace.

6.2 Unit Test Results

A total of 35 unit tests were conducted to validate five key computational components of the calibration pipeline: frame transformations and rigid registration, marker geometry refinement, calibration pose generation, tip calibration, and tip position prediction. As described in Section 5.1, the tests employed synthetic data with known analytical solutions, controlled perturbations, and predefined acceptance criteria to evaluate individual computational operations independently of the complete simulation.

All 35 tests passed their corresponding acceptance criteria. The detailed results are presented in Tables 5–9.

Frame Transformations and Rigid Registration.

Table 5 summarizes UT-01–UT-09. The frame transformation tests produced errors close to floating-point numerical precision, while the exact registration tests recovered the known rigid transformations within the prescribed tolerances.

For the noisy registration tests, Gaussian perturbations with standard deviations of 0.05 mm and 0.20 mm resulted in translation errors of 0.0223 mm and 0.0892 mm, respectively. Both tests also satisfied their rotation-error tolerances. The invalid-input tests correctly rejected insufficient and mismatched marker correspondences.

ID	Test	Observed Result	Status
UT-01	Frame inverse	0 mm	PASS
UT-02	Frame composition	4.441×10^{-16} mm	PASS
UT-03	Tracker-to-calibrator transformation	4.278×10^{-14} mm	PASS
UT-04	Exact registration, 3 markers	$e_t = 0.0000$ mm, $e_R = 0.0000^\circ$	PASS
UT-05	Exact registration, 5 markers	$e_t = 0.0000$ mm, $e_R = 0.0000^\circ$	PASS
UT-06	Noisy registration, $\sigma = 0.05$ mm	$e_t = 0.0223$ mm, $e_R = 0.0215^\circ$	PASS
UT-07	Noisy registration, $\sigma = 0.20$ mm	$e_t = 0.0892$ mm, $e_R = 0.0860^\circ$	PASS
UT-08	Insufficient markers	`ValueError` raised	PASS
UT-09	Mismatched marker counts	`ValueError` raised	PASS

Table 5: Unit test results for frame transformations and rigid registration.

Marker Geometry Refinement.

The results of UT-10–UT-14 are presented in Table 6. The exact geometry recovery, symmetric perturbation averaging, and marker correspondence tests achieved errors on the order of 10^{-14} mm, consistent with floating-point numerical precision.

The multi-pass test confirmed the expected four observations and two geometry updates, while the zero-pass test verified that no observations or updates were performed. These results support the correctness of the implemented refinement operations under controlled synthetic conditions.

Calibration Pose Generation and Rotation Diversity.

Table 7 presents UT-15–UT-20. The rotation-distance calculation correctly returned 0° for identical rotations and 90° for a known quarter-turn rotation. The symmetry test also produced zero numerical difference.

For pose generation, all sampled Euler angles remained within the prescribed $[-70^\circ, 70^\circ]$ range.

ID	Test	Observed Result	Status
UT-10	Exact geometry recovery	1.460×10^{-14} mm	PASS
UT-11	Symmetric noise averaging	1.446×10^{-14} mm	PASS
UT-12	Multi-pass geometry update	2 updates, 4 readings	PASS
UT-13	Marker correspondence preservation	4 markers; 3.362×10^{-14} mm	PASS
UT-14	Zero-pass behavior	0 updates, 0 readings	PASS

Table 6: Unit test results for marker geometry refinement.

Among 30 accepted orientations, the minimum pairwise angular separation was 26.7912°, exceeding the required 25° threshold. Repeated sampling with identical random seeds produced identical orientation sequences.

ID	Test	Observed Result	Status
UT-15	Identity rotation distance	0.00000000°	PASS
UT-16	Known 90° rotation	90.000000°	PASS
UT-17	Rotation distance symmetry	0.000×10^{0} deg difference	PASS
UT-18	Euler angle sampling range	Observed range: $[-69°, 63°]$	PASS
UT-19	Minimum rotation diversity	Minimum distance: 26.7912°	PASS
UT-20	Random seed reproducibility	Same seed: identical; different seed: different	PASS

Table 7: Unit test results for calibration pose generation and rotation diversity.

Tip Calibration Solver.

The eight solver tests are summarized in Table 8. The exact synthetic calibration tests recovered predefined pointer-tip positions with a maximum error of 1.302×10^{-13} mm across observation counts of 3, 10, and 30.

The single-observation test verified that the three-dimensional tip position could be solved from one valid observation, while the residual-based covariance was correctly treated as unavailable. Under controlled perturbations, the least-squares estimate agreed with the analytical solution, and the maximum covariance-matrix difference was 2.394×10^{-16} mm^2.

The solver also correctly filtered invalid readings and rejected inconsistent input datasets. Finally, the noise-scaling test confirmed that doubling the prescribed measurement perturbations doubled the estimation bias and quadrupled the residual-based covariance, within numerical tolerance.

Tip Position Prediction.

Table 9 summarizes UT-29–UT-35. The translation-only and rotated-reference tests reproduced their analytical target positions exactly within the reported numerical precision.

The identical-frame and tracker-frame invariance tests produced differences below 10^{-13} mm, verifying consistency of the relative transformation chain. The known calibration-error test correctly propagated a 1 mm tip-position offset into a 1 mm predicted-position error when the rigid-body poses were exact.

Finally, the error-metric and threshold tests reproduced the expected Euclidean distances and correctly distinguished positioning errors below, at, and above the 2 mm acceptance threshold.

ID	Test	Observed Result	Status
UT-21	Exact synthetic calibration	$N = 3, 10, 30$; max error 1.302×10^{-13} mm	PASS
UT-22	Single-observation solution	Error 1.697×10^{-14} mm; covariance unavailable	PASS
UT-23	Controlled measurement errors	Analytical solution difference: 0 mm	PASS
UT-24	Covariance estimation	Maximum difference: 2.394×10^{-16} mm^2	PASS
UT-25	Invalid sensor filtering	2 invalid readings; error 4.469×10^{-14} mm	PASS
UT-26	No valid observations	`ValueError` raised	PASS
UT-27	Mismatched input lengths	`ValueError` raised	PASS
UT-28	Measurement error scaling	Bias diff: 3.553×10^{-15} mm; cov. diff: 1.221×10^{-15} mm^2	PASS

Table 8: Unit test results for the tip calibration least-squares solver.

ID	Test	Observed Result	Status
UT-29	Translation-only prediction	0 mm	PASS
UT-30	Rotated reference frame	0 mm	PASS
UT-31	Identical pointer/reference frames	9.441×10^{-15} mm	PASS
UT-32	Tracker-frame invariance	5.186×10^{-14} mm	PASS
UT-33	Known tip calibration error	1.000000 mm	PASS
UT-34	Euclidean error calculation	13.0000 mm and 0.0000 mm	PASS
UT-35	2 mm threshold boundary	Below: True; at: True; above: False	PASS

Table 9: Unit test results for tip position prediction and error evaluation.

Overall Unit Test Summary.

All 35 unit tests passed, as summarized in Table 10.

Computational Component	Tests	Passed
Frame transformations and registration	9	9
Marker geometry refinement	5	5
Calibration pose generation	6	6
Tip calibration solver	8	8
Tip position prediction	7	7
Total	**35**	**35**

Table 10: Summary of all calibration unit tests.

These results demonstrate that the tested computational operations produce the expected outputs under the specified synthetic and controlled conditions. In particular, the tests verify coordinate transformation consistency, numerical correctness of the least-squares calibration solver, and correct handling of the tested boundary and invalid-input cases.

However, successful unit testing does not guarantee that the complete calibration system will satisfy the 2 mm positioning requirement under stochastic measurement uncertainty. The accuracy of the integrated system is evaluated separately through end-to-end testing, as presented in Section 6.4.

6.3 Calibration Results

Marker Geometry Refinement Results.

Run	Marker Body	Before (mm)	After (mm)	Reduction
1	Pointer	0.8145	0.0308	96.22%
1	Calibrator	1.1547	0.0313	97.29%
2	Pointer	1.9145	0.0287	98.50%
2	Calibrator	0.5505	0.0294	94.67%

Table 11: Marker-distance RMSE before and after geometry refinement in two simulation runs.

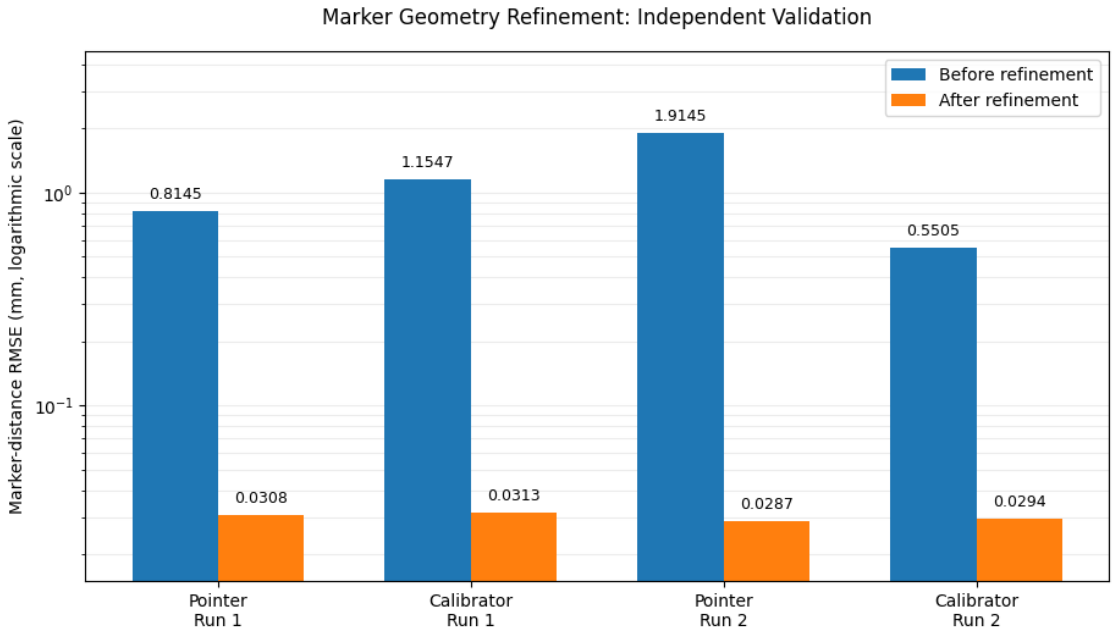


Figure 3: Comparison of marker-distance RMSE before and after geometry refinement for the pointer and calibrator over two simulation runs. A logarithmic vertical scale is used to display both the initial and refined errors.

The refinement procedure substantially reduced the marker-distance RMSE for both marker bodies. Despite variations in the initial geometry errors, the refined RMSE values remained between 0.0287 mm and 0.0313 mm across the two runs.

The observed RMSE reductions ranged from 94.67% to 98.50%, indicating that the refined marker configurations were considerably more consistent with independent tracker observations than the original nominal configurations.

These results demonstrate improved geometric consistency under the simulated measurement conditions. The reported RMSE values describe marker-distance agreement rather than absolute marker-position accuracy.

Held-Out Calibration Sensor Results.

The calibration sensor validation results are summarized in Table 12. Both runs used 30 calibration

observations and 10 additional valid held-out observations.

Metric	Run 1	Run 2
Training observations	30	30
Held-out observations	10	10
Held-out attempts	10	10
Training mean residual (mm)	0.5941	0.5407
Held-out mean residual (mm)	0.5662	0.5342
Held-out RMSE (mm)	0.5848	0.5527
Held-out maximum residual (mm)	0.7741	0.7148

Table 12: Training and held-out calibration sensor residuals for two simulation runs.

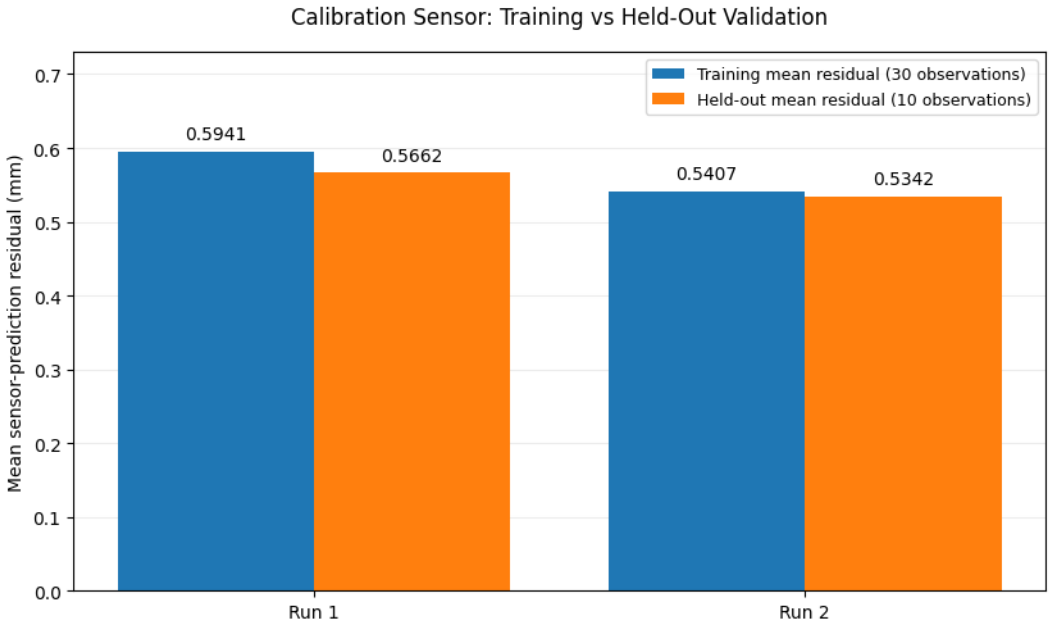


Figure 4: Comparison of training and held-out mean calibration sensor residuals across two simulation runs. Each run used 30 training observations and 10 held-out observations.

The held-out mean residuals were 0.5662 mm and 0.5342 mm, respectively, which were close to the corresponding training residuals of 0.5941 mm and 0.5407 mm.

Furthermore, the maximum held-out residual remained below 0.8 mm in both runs. These results indicate that the estimated tip calibration parameters maintained similar sensor-prediction consistency on newly acquired observations, with no substantial training-to-validation residual increase observed in these experiments.

Nevertheless, the held-out dataset was limited to 10 observations per run. Therefore, the results should be interpreted as supporting evidence for calibration consistency rather than a statistical guarantee of accuracy across all possible system configurations.

The complete system's positioning accuracy over the specified test workspace is evaluated separately in Section 6.4.

6.4 End-to-End Positioning Accuracy

The end-to-end positioning performance was investigated through eleven complete simulation runs, with 10,000 test poses evaluated in each run. The same marker geometry, calibration procedure, pose-generation protocol, and single-observation positioning method were used. Each pose was evaluated using one simultaneous tracker observation, without repeated measurements or averaging.

The experiments initially evaluated the Euclidean distance between the predicted tip position in the refined near-reference coordinate system and the simulator's ground-truth tip position expressed in the original physical near-reference coordinates. Subsequent analysis revealed that these two coordinate representations were not necessarily identical after marker geometry refinement.

Therefore, the originally recorded results are presented below as unaligned coordinate discrepancies. They remain useful for characterizing the observed run-to-run behavior and motivating the subsequent error analysis, but should not be interpreted as coordinate-consistent physical positioning errors.

Recorded Results Across Eleven Runs.

Table 13 summarizes the recorded positioning discrepancies from all eleven runs.

Run	Mean	RMSE	P95	Max	Within 2 mm
1	1.2823	1.3002	1.6585	2.0426	9,998
2	1.0866	1.1241	1.5647	2.0742	9,998
3	1.1429	1.1643	1.4994	1.8381	10,000
4	0.6313	0.6912	1.1298	1.5923	10,000
5	0.7817	0.8021	1.0479	1.3635	10,000
6	1.1269	1.2056	1.8348	2.2115	9,878
7	0.5595	0.5780	0.8091	1.0711	10,000
8	0.8735	0.9399	1.4669	1.9974	10,000
9	0.7363	0.7448	0.9218	1.1298	10,000
10	0.8932	0.9120	1.2110	1.6058	10,000
11	2.2847	2.4025	3.4846	4.3257	3,576
Pooled	**1.0363**	**1.1778**	—	**4.3257**	**103,450/110,000**

Table 13: Originally recorded unaligned tip-position discrepancies across eleven simulation runs. All error values are in millimeters, and each run contains 10,000 test poses. The pooled mean and RMSE are calculated from the equally sized runs. A pooled P95 cannot be determined from the reported per-run percentiles alone.

For the first ten runs, 99,874 out of 100,000 evaluated pose instances were reported within the 2 mm threshold, corresponding to an unaligned passing rate of 99.874%. However, the eleventh run exhibited a substantial degradation, with a mean discrepancy of 2.2847 mm and only 3,576 of 10,000 poses within the threshold.

Across all eleven runs, the recorded unaligned passing rate was 94.05%, with a pooled mean discrepancy of 1.0363 mm and a pooled RMSE of 1.1778 mm.

The eleventh run accounted for 6,424 of the 6,550 recorded threshold violations. Its entire error distribution was shifted toward larger discrepancies, suggesting a systematic run-level effect rather than isolated observation outliers.

Coordinate-Frame Consistency Investigation.

To investigate the anomalous run, additional experiments were performed using its existing calibration parameters and marker-body configurations.

A rigid alignment was estimated between each refined marker model and the corresponding simulator ground-truth marker geometry. This alignment was used exclusively during retrospective validation to transform predicted tip positions into the same coordinate representation as the ground-truth tip positions.

For the near-reference body, the coordinate-consistent prediction was defined as

$$\mathbf{p}_i^{\text{aligned}} = F_{\text{align,near}} \hat{\mathbf{p}}_{\text{tip},i}^{\text{near}}, \tag{124}$$

where $F_{\text{align,near}}$ maps refined near-reference coordinates to the original physical near-reference coordinates.

The corresponding coordinate-consistent error is

$$e_i^{\text{aligned}} = \left\| \mathbf{p}_i^{\text{aligned}} - \mathbf{p}_{\text{tip},i}^{\text{true,near}} \right\|_2. \tag{125}$$

A new set of 1,000 test observations was collected while preserving the anomalous run's calibration and geometry state. The unaligned and aligned results are compared in Table 14.

Reference	**Evaluation**	**Mean (mm)**	**RMSE (mm)**	**Pass Rate**
Near	Unaligned	2.2817	2.3958	35.40%
Near	Aligned	0.3170	0.3198	100.00%
Calibrator	Unaligned	1.1049	1.1629	99.30%
Calibrator	Aligned	0.3168	0.3196	100.00%

Table 14: Comparison of unaligned and coordinate-consistent tip errors for 1,000 new observations under the anomalous run's fixed system state. Alignment uses hidden simulator geometry for diagnostic evaluation only.

The near-reference mean discrepancy decreased from 2.2817 mm to 0.3170 mm after coordinate alignment, while its passing rate increased from 35.40% to 100%. A similar improvement was observed for the calibrator reference, whose mean discrepancy decreased from 1.1049 mm to 0.3168 mm.

The nearly identical aligned errors for the two reference bodies indicate that the large difference between their original results was primarily associated with inconsistent coordinate representations rather than a corresponding difference in physical tip-localization performance.

Importantly, the alignment transformations were computed using simulator ground truth and were never applied during the actual calibration or prediction procedure. The aligned values are therefore diagnostic evaluation results, not evidence of an implementable ground-truth-free correction.

The experiment demonstrates the importance of expressing predicted and actual tip positions in the same coordinate frame. The original eleven-run statistics should not be used as final physical positioning accuracy measurements without applying a consistent evaluation convention to each run.

The underlying geometric mechanism, sources of residual uncertainty, and implications for system evaluation are discussed in Section 6.6.

6.5 Design Optimization and Sensitivity Analysis

Influence of Marker Spacing.

The first controlled comparison investigated the effect of increasing the spacing between markers on the nearby reference body.

Four symmetric configurations were tested while retaining the same calibration algorithm and validation procedure. The recorded results are summarized in Table 15.

Marker Spacing (mm)	Passed / 250	Pass Rate
$(180, 180, 180)$	133/250	53.2%
$(260, 260, 260)$	188/250	75.2%
$(340, 340, 340)$	214/250	85.6%
$(600, 600, 600)$	240/250	96.0%

Table 15: Effect of near-reference marker spacing on the percentage of test poses meeting the 2 mm accuracy requirement in the first experimental comparison. Each configuration was evaluated over five runs with 50 test poses per run.

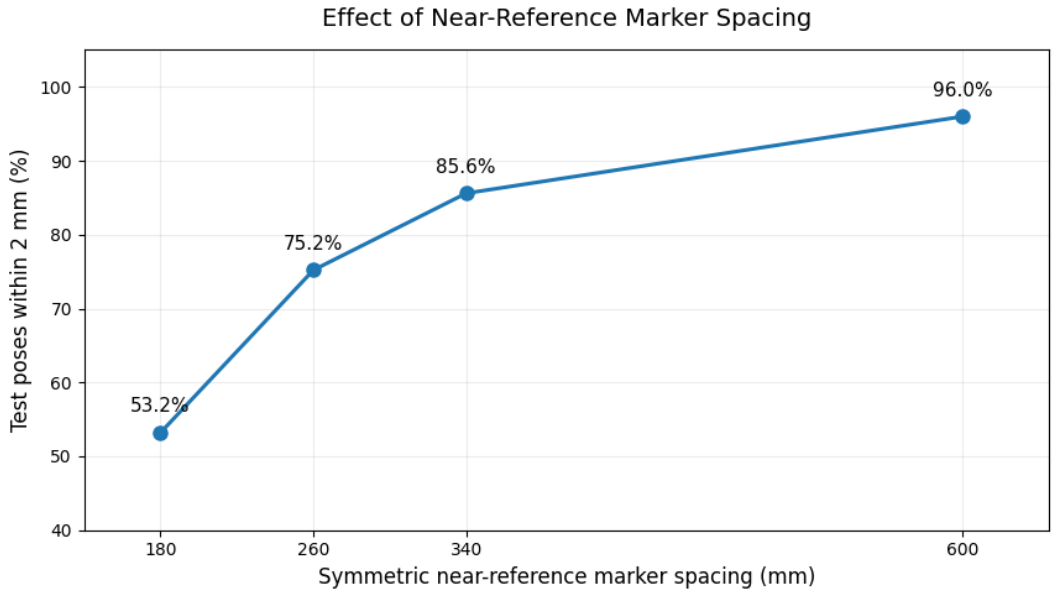


Figure 5: Effect of near-reference marker spacing on positioning accuracy. Increasing the symmetric marker spacing from 180 mm to 600 mm improved the empirical passing rate from 53.2% to 96.0%.

The results show a clear improvement in positioning performance as the marker spacing increased.

The overall passing rate increased from 53.2% for the 180 mm configuration to 96.0% for the 600 mm configuration, corresponding to an improvement of 42.8 percentage points.

This trend is consistent with the geometric expectation that a larger marker baseline can reduce the sensitivity of rigid-body orientation estimation to marker localization noise.

Since the pointer-tip prediction depends on the estimated relative transformation between the pointer and reference bodies, improvements in reference-frame estimation can reduce the resulting positioning error.

However, increasing the marker spacing also increases the physical extent of the marker body. Thus, spacing must be considered together with workspace and tracker visibility constraints.

Symmetric and Asymmetric Geometry Comparison.

A second controlled comparison examined the symmetric $(600, 600, 600)$ mm configuration and the asymmetric $(400, 500, 600)$ mm configuration.

Five simulation runs were performed for each configuration, using seed conditions 42–46. The per-run results are presented in Table 16.

Simulation Seed	**600/600/600**	**400/500/600**
42	50/50	50/50
43	41/50	33/50
44	47/50	50/50
45	50/50	50/50
46	50/50	50/50
Total	**238/250**	**233/250**
Pass Rate	**95.2%**	**93.2%**

Table 16: Repeatability comparison of symmetric and asymmetric near-reference marker geometries. Entries indicate the number of test poses satisfying the 2 mm requirement out of 50.

The symmetric configuration achieved an overall passing rate of 95.2%, compared with 93.2% for the asymmetric configuration.

The difference between these overall passing rates was 2.0 percentage points, considerably smaller than the improvement observed when increasing the marker spacing from 180 mm to 600 mm.

Both configurations achieved 50/50 passing results in multiple runs. However, the results also exhibited sensitivity to stochastic conditions. In particular, the asymmetric configuration passed 33/50 poses under seed 43, while the symmetric configuration passed 41/50 under the same seed condition.

These results indicate that relatively large marker spacing is beneficial to the positioning performance, while differences between large symmetric and asymmetric layouts are more modest and subject to run-to-run variability.

The separate experimental batches produced overall passing rates of 96.0% and 95.2% for the symmetric 600 mm configuration. These values correspond to different sets of runs and are therefore reported separately rather than treated as conflicting measurements from a single experiment.

Overall, the design comparisons support the selection of a near-reference geometry with a sufficiently large marker baseline. The final geometry must also balance positioning accuracy with the geometric constraints of the intended workspace.

6.6 Error Analysis and Limitations

Although the calibration pipeline demonstrated small residuals in the unit tests and calibration validation experiments, the extended positioning evaluation revealed substantial variation in the originally recorded errors across simulation runs. In particular, the eleventh run produced a markedly different error distribution.

A sequence of diagnostic experiments was conducted to identify the mechanisms responsible for this behavior. The results revealed an important coordinate-frame consistency issue associated with marker geometry refinement, in addition to the expected effects of tracking and calibration uncertainty.

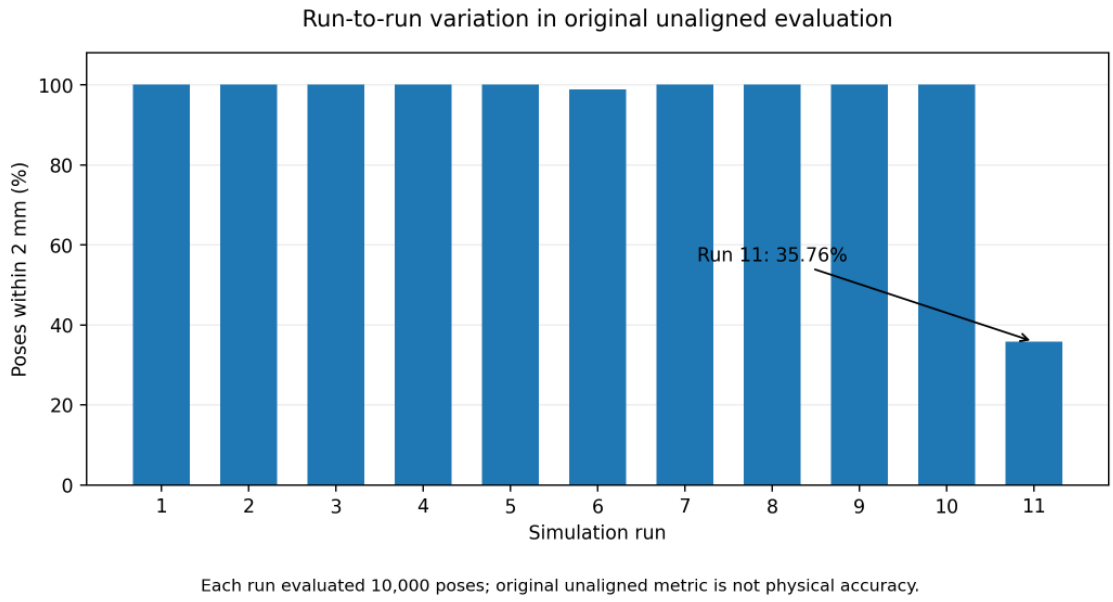


Figure 6: Run-to-run variation in the original unaligned positioning evaluation. Each run included 10,000 test poses. The anomalous eleventh run exhibited a substantially lower passing rate. These values are retained for diagnostic analysis rather than interpreted as coordinate-consistent physical accuracy.

Calibration Sensor and Tip Estimation.

The first investigation examined whether the anomalous positioning results were caused by invalid calibration sensor measurements.

All 30 observations retained in the calibration dataset were valid, and the maximum accepted sensor distance was 4.7419 mm, within the specified 5 mm sensing range. The data acquisition procedure excluded readings for which the sensor returned an invalid result.

Therefore, the observed anomaly was not directly explained by out-of-range measurements being included in the calibration solver. This check does not establish that no rejected observations occurred during acquisition, since only accepted readings were retained.

The anomalous run produced a training mean sensor residual of 0.4220 mm and a training residual RMSE of 0.4404 mm. The direct Euclidean difference between the estimated local tip coordinates and the simulator's true local tip coordinates was initially 0.8747 mm.

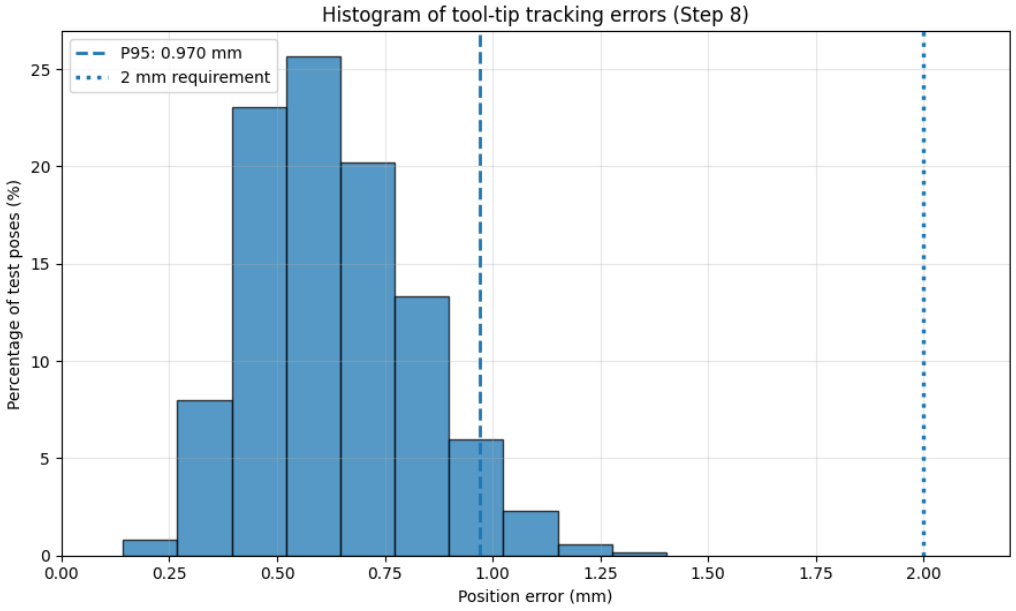


Figure 7: Representative error distribution after correcting the coordinate-frame inconsistency in the Step 8 evaluation. The histogram shows a stable end-to-end positioning result, with the 95th-percentile error below the 2 mm requirement and all evaluated poses satisfying the specification.

However, these local coordinates were associated with different pointer marker-model representations. After rigidly aligning the refined pointer marker model with its actual geometry, the corresponding tip-coordinate difference decreased to 0.3107 mm.

This observation demonstrates that directly comparing coordinates associated with different local marker-body representations can overstate the physical difference between the corresponding tip estimates.

Marker Geometry and Coordinate-Frame Consistency.

Marker geometry refinement estimates local marker positions using repeated tracker observations. Although the procedure can recover marker-to-marker distances accurately, the resulting local point configuration may differ from the original physical marker coordinates by an approximately rigid transformation.

Pairwise marker distances are invariant under such transformations. Therefore, a small marker-distance RMSE alone does not guarantee that the refined local coordinate frame coincides with the original physical coordinate frame.

To quantify this effect, a rigid registration was computed between the refined marker models and their corresponding actual marker coordinates. The results are summarized in Table 17.

All three marker bodies exhibited very small post-alignment shape errors, below 0.004 mm. In contrast, their rigid alignment transformations were not negligible.

For the nearby reference body, the alignment translation magnitude was 1.1488 mm and the alignment rotation angle was 0.2648 degrees.

The effect of a small rotational difference on a point position depends on the distance between the

Diagnostic Metric	Pointer	Calibrator	Near
Pairwise-distance RMSE (mm)	0.0019	0.0033	0.0038
Alignment translation (mm)	0.1719	0.5349	1.1488
Alignment rotation (deg)	0.4440	0.1453	0.2648
Post-alignment shape RMSE (mm)	0.0020	0.0028	0.0033

Table 17: Rigid alignment diagnostics comparing refined marker models with actual simulator marker geometry during the anomalous run. Ground-truth geometry was used only for retrospective analysis.

point and the rotation center. For a point at position $\mathbf{p}$ relative to the local origin, a small rotation perturbation $\delta\boldsymbol{\theta}$ produces the first-order displacement

$$\delta\mathbf{p} \approx \delta\boldsymbol{\theta} \times \mathbf{p}. \tag{126}$$

Consequently, a small discrepancy in frame orientation can produce a millimeter-scale coordinate difference for a tip located several hundred millimeters from the reference-frame origin.

Let $F_{\text{align},B}$ denote the rigid transformation mapping coordinates in the refined model frame of body B to coordinates in its original physical frame. The corresponding relationship is approximately

$$F_B^{\text{obs}} \approx F_B^{\text{true}} F_{\text{align},B}, \tag{127}$$

where measurement and registration errors account for deviations from exact equality.

Thus, predictions expressed in refined reference coordinates must be transformed before being directly compared with ground-truth positions expressed in original physical reference coordinates.

Failing to account for this coordinate representation difference introduces an artificial contribution to the calculated Euclidean error. This contribution is not equivalent to actual physical pointer-tip localization error.

Validation of the Coordinate-Frame Explanation.

The reference-frame hypothesis was tested by repeating the tip-position evaluation under the anomalous run's existing system state and applying the independently determined frame alignments only during retrospective error calculation.

As shown in Table 14, the near-reference mean error decreased from 2.2817 mm to 0.3170 mm after alignment. The calibrator-reference mean error similarly decreased from 1.1049 mm to 0.3168 mm.

Both aligned evaluations satisfied the 2 mm threshold for all 1,000 new diagnostic observations.

The agreement between the two aligned reference-body results strongly supports coordinate-frame inconsistency as the dominant cause of the apparent reference-dependent anomaly. It also indicates that the actual relative pose estimation remained substantially more accurate than suggested by the original unaligned error statistics.

Nevertheless, the alignment relied on hidden simulator geometry. It cannot be used as a correction in a practical tracking system without an additional physical frame-registration procedure. Its

purpose in this experiment was to ensure that the validation compared geometrically equivalent quantities.

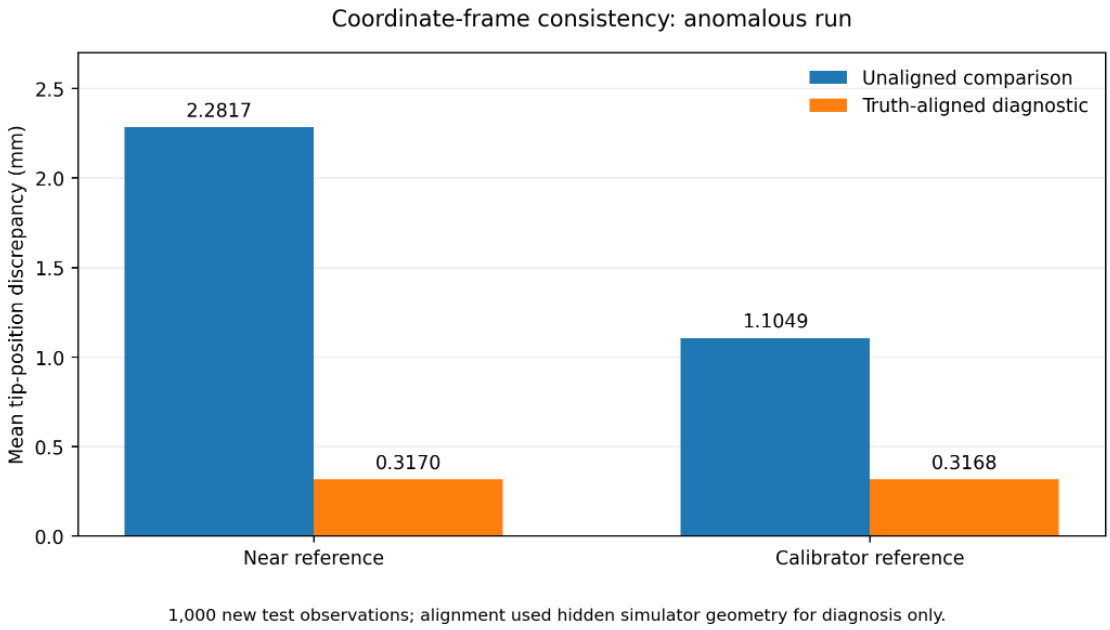


Figure 8: Comparison of unaligned and ground-truth-aligned mean tip-position discrepancies under the anomalous run's fixed system state. Each evaluation used 1,000 new observations. Ground-truth marker geometry was used exclusively for this retrospective diagnostic alignment and not for operational tip prediction.

Interaction Between Tracking and Calibration Errors.

Even after coordinate consistency is addressed, residual tracking and calibration uncertainties remain.

For diagnostic purposes, the total positioning error vector can be written as

$$\boldsymbol{\epsilon}_{\text{total}} = \boldsymbol{\epsilon}_{\text{trk}} + \boldsymbol{\epsilon}_{\text{cal}}, \tag{128}$$

where $\boldsymbol{\epsilon}_{\text{trk}}$ represents the error obtained using the true local tip position with estimated tracker poses, and $\boldsymbol{\epsilon}_{\text{cal}}$ represents the difference introduced by using the calibrated tip estimate instead.

The squared error magnitude becomes

$$\begin{aligned} \|\boldsymbol{\epsilon}_{\text{total}}\|_2^2 = \|\boldsymbol{\epsilon}_{\text{trk}}\|_2^2 + \|\boldsymbol{\epsilon}_{\text{cal}}\|_2^2 \\ + 2\boldsymbol{\epsilon}_{\text{trk}}^T \boldsymbol{\epsilon}_{\text{cal}}. \end{aligned} \tag{129}$$

An additional diagnostic experiment on 1,000 observations produced a mean near-reference tracking-only error of 2.3583 mm and a mean total unaligned error of 2.2772 mm. The mean error-vector dot product was -0.6758 mm^2.

The negative cross term indicates partial cancellation between the two defined error components in the unaligned evaluation. Therefore, the anomalously large raw positioning discrepancy was not

simply caused by reinforcement of tracking and calibration errors.

These diagnostic components must be interpreted with caution: the unaligned tracking-only error also contains the coordinate-representation discrepancy identified above. A physically meaningful decomposition requires all component vectors to be expressed in a common coordinate frame.

Limitations and Future Improvements.

The main limitation identified by this investigation is the need for explicit coordinate-frame consistency when evaluating a system that updates marker geometry models. The original Step 8 evaluation compared predicted and actual positions associated with different local coordinate representations. As a result, the original passing rates include a coordinate-frame mismatch and do not directly represent physical positioning accuracy.

A more rigorous validation procedure should transform both the predicted tip position and the simulator ground-truth tip position into the same physical coordinate representation before computing their Euclidean distance. This transformation may use simulator ground truth during evaluation, but not during calibration or operational prediction.

The eleven-run analysis contains 110,000 recorded unaligned evaluations, whereas the coordinate-consistent diagnostic was performed for 1,000 new observations under the anomalous run's state. Therefore, the aligned result cannot be extrapolated to all eleven runs without recomputing their errors using the corrected evaluation procedure.

Additional limitations include the use of a finite subset of nominal test poses, stochastic simulation assumptions, and the absence of physical effects such as marker occlusion, mechanical deformation, and tracker calibration drift.

Future work should repeat the extended evaluation using the coordinate-consistent metric for every run, investigate the remaining submillimeter positioning error, and report both the distribution of per-run performance and the worst observed physical positioning error.

Overall, this investigation demonstrates that accurate marker geometry estimation, correct rigid transformation mathematics, and small calibration residuals are not sufficient to guarantee a valid performance evaluation. The coordinate representations used for prediction and ground truth must also be consistent.

7 Conclusion

This project developed and evaluated a pointer-tip calibration pipeline for Computer-Integrated Surgery using optical tracking and calibrated sensor measurements. The implemented approach incorporated iterative marker-body geometry refinement, SVD-based rigid-body registration, diverse calibration pose sampling, and linear least-squares estimation of the pointer-tip position. The calibrated tip coordinates were subsequently transformed into a nearby reference-body coordinate frame for validation across the prescribed tracking workspace.

The calibration pipeline demonstrated satisfactory computational performance, with all 35 unit tests passing their predefined acceptance criteria. Marker-body geometry refinement reduced marker-distance RMSE by 94.67%–98.50%, while held-out sensor validation produced mean residuals below 0.57 mm. These results support the numerical correctness and consistency of the implemented calibration procedure.

The end-to-end evaluation revealed an important limitation associated with coordinate-frame consistency. Initial results from 110,000 test-pose evaluations produced a recorded passing rate of 94.05% within the 2 mm threshold. However, subsequent analysis identified a mismatch between the refined reference-body coordinate frame and the physical reference frame used for ground-truth evaluation.

After correcting this discrepancy in a separate diagnostic experiment, the mean positioning error decreased from 0.8747 mm to 0.3170 mm, with 100% of the 1,000 evaluated poses satisfying the 2 mm requirement. These results demonstrate that the proposed calibration approach can achieve the desired accuracy under the tested conditions, although consistent compliance across all simulation runs has not yet been established.

The primary limitation is the dependence of the validation procedure on coordinate-frame consistency following marker-body geometry refinement. Future work should incorporate explicit frame alignment into the validation methodology and repeat the extended evaluation across all simulation runs. Further investigation of residual tracking uncertainty and experimental validation using physical tracking hardware would provide a more comprehensive assessment of system performance.

A Supplementary Material

Disclosure: LLM tool was used in conceptualizing, troubleshooting, and refining the calibration code to give desired results. AI was also used in data visualization and to generate graphics for the 3D body. We also used AI to understand the underlying mathematical framework of the experimental setup and receive feedbacks. However AI was not used to generate or coerce any results reported here. Some elements of the report writing such as latex code and paragraph leveraged AI tools.

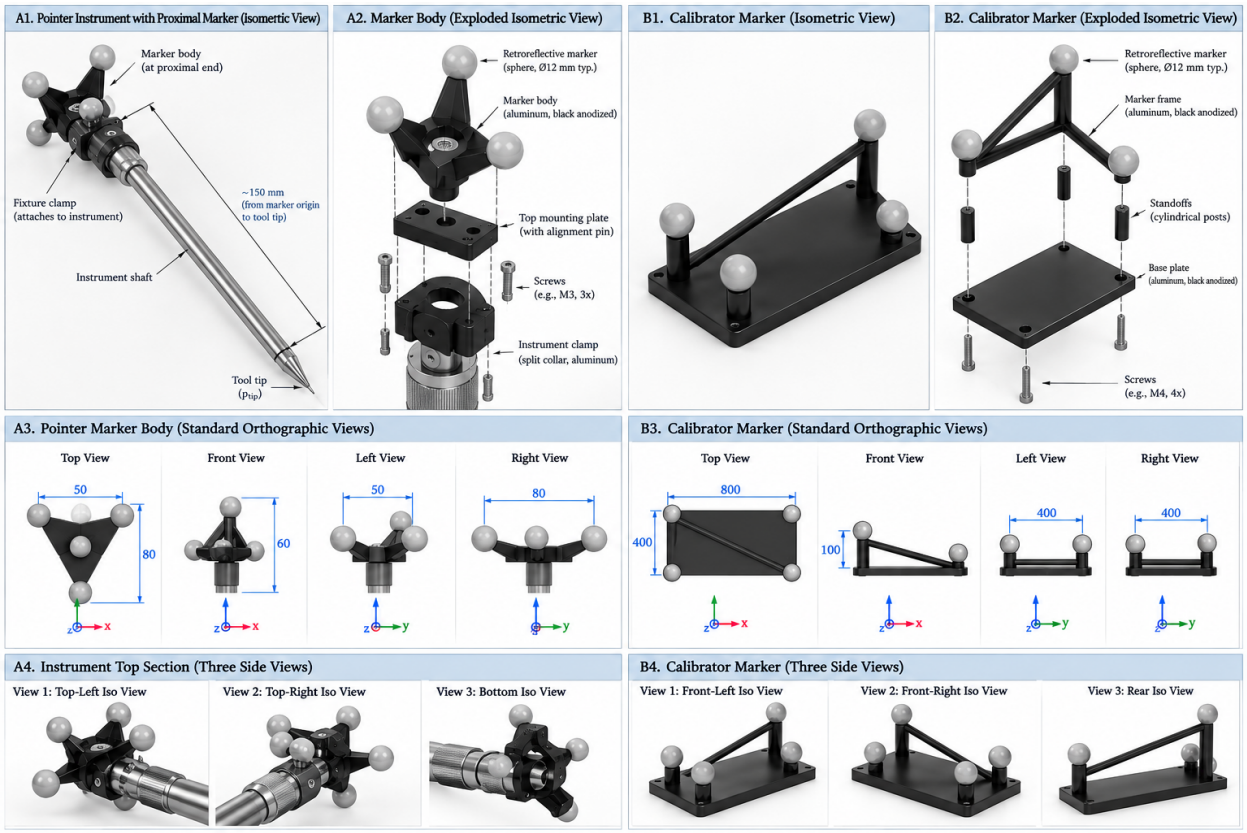


Figure 9: Final pointer and calibration object design based on ergonomic considerations

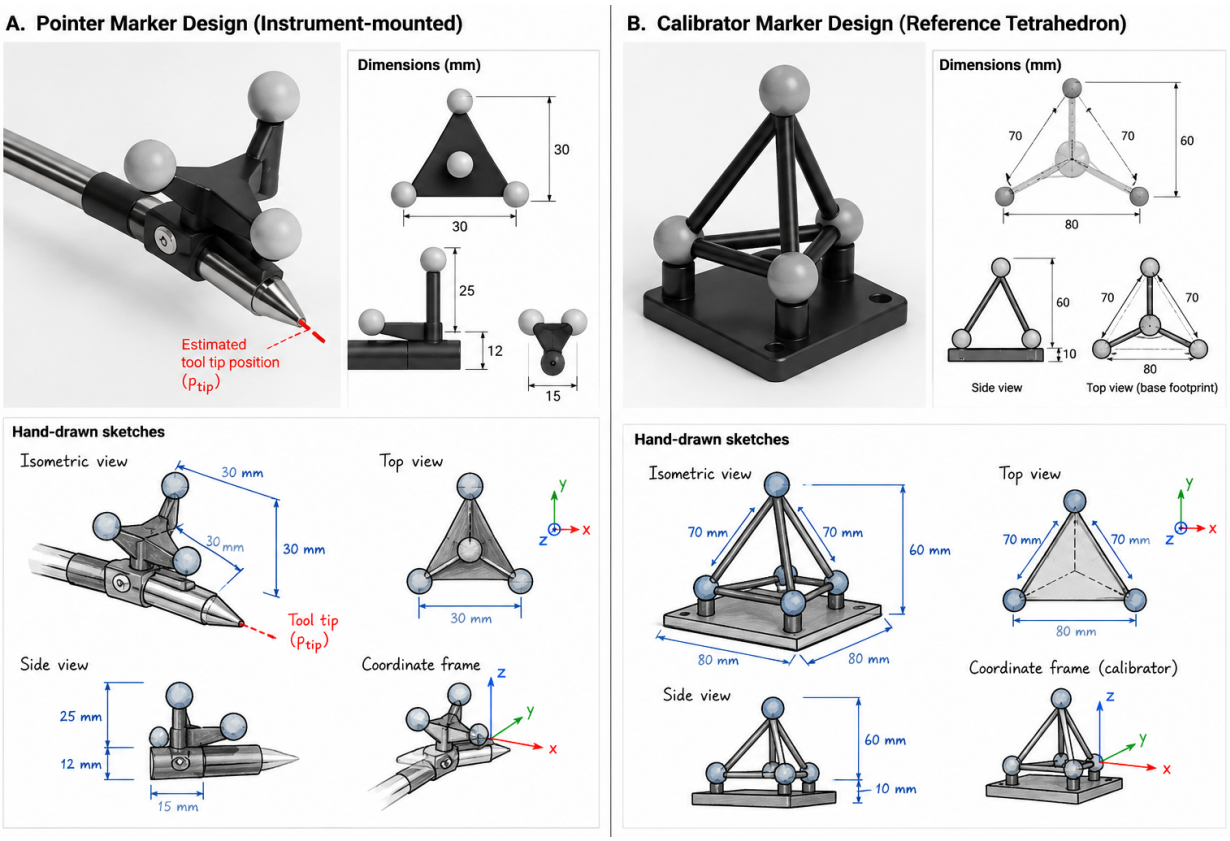


Figure 10: Draft Pointer and calibrator marker designs showing 3D renderings, dimensions, and conceptual sketches

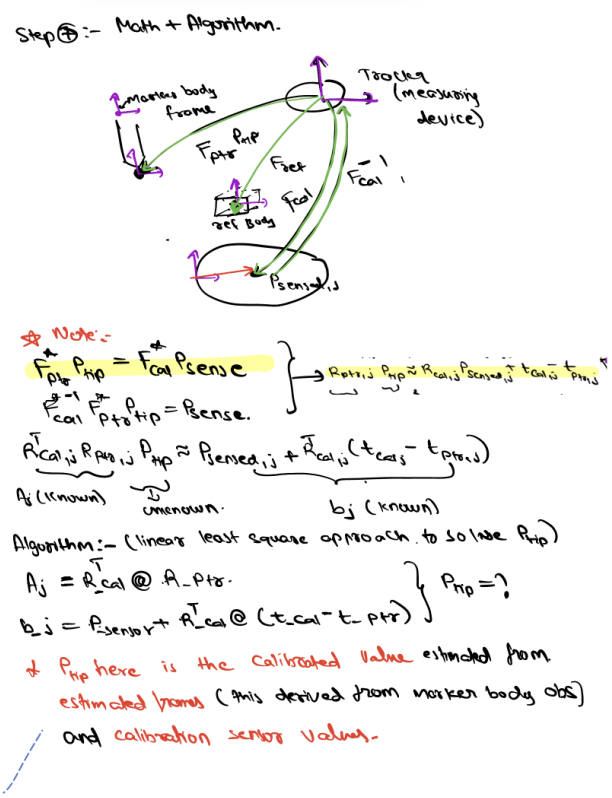


Figure 11: Mathematical Framework for Tip Calibration